Capacitance

Capacitance:

→ It is capacity of a metal or conductor to store charge.

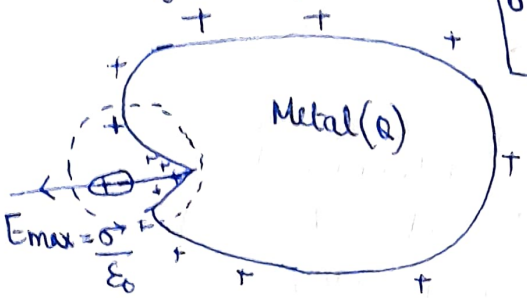
→ If a metal has → less capacity to store charge ⇒ less capacitance
→ More capacity to store charge ⇒ More capacitance

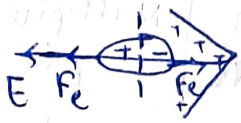
→ Metal/Conductor can store charge only upto a maximum level.

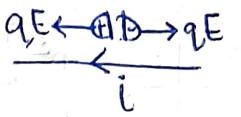
Q. We cannot give infinite charge to a metal or a conductor. Why?

A. Insulating medium/Dielectric

$\sigma \propto \frac{1}{R_c}$





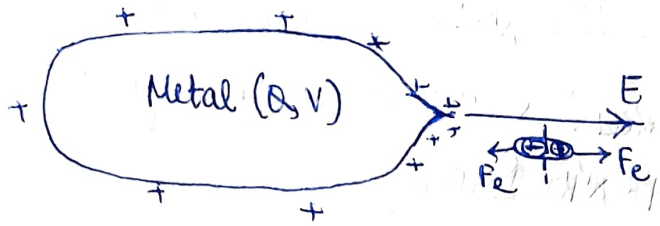


As Q↑, electric field↑, due to which force on ~~dielect~~ molecules of the medium also↑. For a value of E, molecule starts to break now, -ve charge gets deposited on metal surface and neutralizes the charge Q, hence not letting it cross a certain limit. A current also develops in the region.

→ Min^m electric field at which dielectric medium starts behaving like a conductor is known as dielectric strength/breakdown electric field.

→ When electric field reaches at the value of dielectric strength, then molecules of dielectric/insulator medium starts to break.

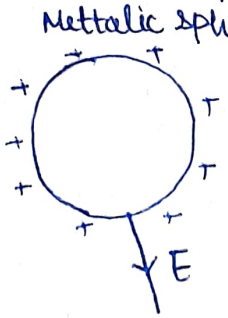
Breakdown Voltage:



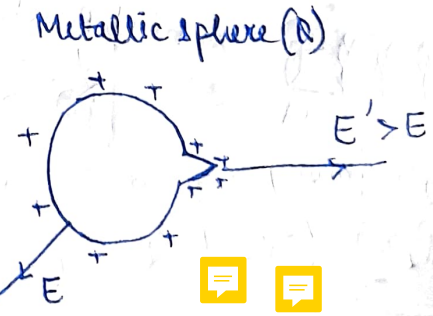
* Potential at which its insulating surrounding starts to behave like a conductor is known as Breakdown voltage $\{V_B\}$.

Ex Which has more capacitance?

(A)



(B)



Same medium

In (B), molecules of medium can break faster as at sharp edge, $E' > E$.

So (A) can store more charge as compared to (B)

∴ (A) has more capacitance.

* Capacitance of a metal depends on:

① Shape, size, geometry of metal

Metal Surface → Flat ⇒ Capacitance ↑
→ Sharp ⇒ Capacitance ↓

② Surrounding Medium

Molecule → stable (strong) ⇒ Capacitance ↑
→ weak ⇒ Capacitance ↓

Capacitance of a conductor:

* $Q \propto V$

$\Rightarrow Q = CV$

* $\boxed{C = \frac{Q}{V} = \text{Capacitance}}$

Note: C does not depends on Q and V. It depends on size and geometry. It is just ratio of Q and V. If $Q \to 2Q$, then $V \to 2V$ so, $C = \frac{2Q}{2V} = \frac{Q}{V}$.

→ Unit = $\frac{\text{Coulomb}}{\text{Volt}}$ = Farad (F)

→ $MF = 10^{-6} F$, $nF = 10^{-9} F$

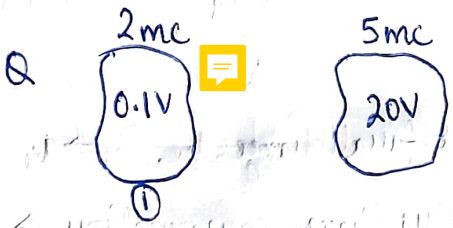
→ Dimension = $\frac{I^2T^2}{ML^2T^{-2}} = [M^{-1}L^{-2}T^4I^2]$

→ Scalar

→ Non-negative.

$C_1 = \frac{5mC}{10V} = 0.5\,nF$ $C_2 = \frac{6mC}{10V} = 0.6mF$

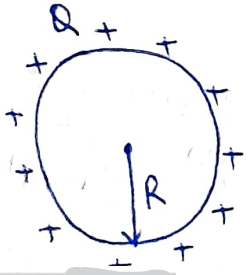
$\boxed{C_1 < C_2}$



$C_1 = \frac{2mC}{0.1V} = 20mF$ $C_2 = \frac{5mc}{20v} = \frac{1}{4}mF$

$\boxed{C_1 > C_2}$

Capacitance of a metallic sphere:

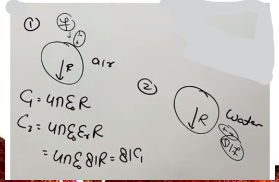


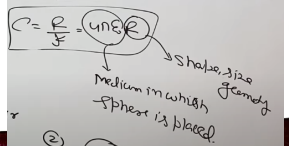
$C = \frac{Q}{V} = \frac{QR}{KQ} = \frac{R}{K}$ $\therefore \boxed{C = \frac{R}{K}}$

$\boxed{C = 4\pi\varepsilon_0 R \text{ (in air)}}$

$\boxed{C = 4\pi\varepsilon_0\varepsilon_r R \text{ (in any other medium)}}$

∴ C depends on radius of sphere and the medium in which it is kept.





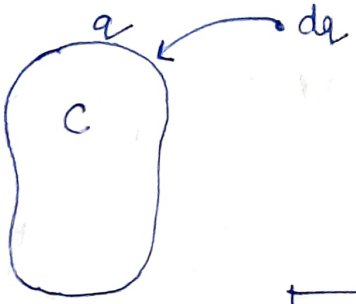
Q Find capacitance of earth.

$$C = \frac{R}{k} = \frac{6400000}{9\times10^9} \text{ Farad}$$

$$\boxed{C = 711\,\mu F}$$

we know the capacitance of a sphere is directly proportional to its radius
the largest radius sphere possible, earth, itself has a capacitance only in micro farads. Thus, we will have to make some arrangements to obtain capacitors of higher order

Energy stored by Conductor/Metal (Self Energy):



$$dW = V\,dq$$

$$dW = \frac{q}{C}\,dq \qquad \left(C = \frac{q}{V}\right)$$

$$W_{ext} = \int dW = \frac{1}{C}\int_0^Q q\,dq = \frac{1}{C}\left(\frac{Q^2}{2}\right)$$

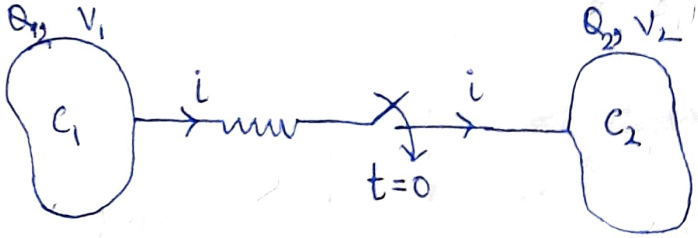
$$\boxed{U_{self} = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV}$$

$$U = \int \frac{1}{2}\varepsilon E^2\,dV$$

Sharing of charges b/w 2 conductors:

2 metals

assuming V1>V2



$$Q_1 = C_1V_1$$
$$Q_2 = C_2V_2$$

Let final charges be $Q_1 \to q_1$ and $Q_2 \to q_2$ and final potential be V_{common}.

Using Charge conservation $\Rightarrow$ $Q_1+Q_2 = q_1+q_2$ — (1)

$\Rightarrow$ $C_1V_1+C_2V_2 = q_1+q_2$

Final potential must be equal $\Rightarrow$ $V_{common} = \frac{q_1}{C_1} = \frac{q_2}{C_2}$

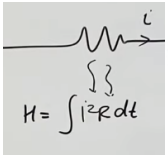
From (1) and (2)

$$\boxed{q_1 = \frac{C_1}{C_1+C_2}(Q_1+Q_2) = \frac{C_1}{C_1+C_2}(C_1V_1+C_2V_2)}$$

$$\Rightarrow \boxed{\frac{q_1}{q_2} = \frac{C_1}{C_2}} \quad — (2)$$

$$\boxed{q_2 = \frac{C_2}{C_1+C_2}(Q_1+Q_2) = \frac{C_2}{C_1+C_2}(C_1V_1+C_2V_2)}$$

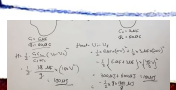
$$\boxed{V_{common} = \frac{q_1}{C_1} = \frac{q_2}{C_2} = \frac{C_1V_1+C_2V_2}{C_1+C_2}}$$



$\circledast$ Heat dissipated $= U_i - U_f$

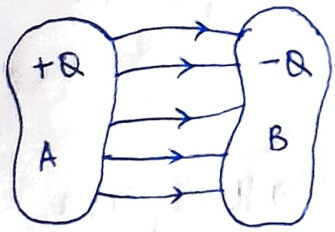
$$= \left(\frac{1}{2}C_1V_1^2+\frac{1}{2}C_2V_2^2\right) - \left(\frac{1}{2}C_1V_{com}^2+\frac{1}{2}C_2V_{com}^2\right)$$

$$\Rightarrow \boxed{\text{Heat} = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1-V_2)^2}$$



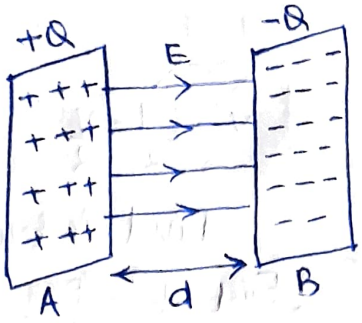
Capacitor / Condenser:

→ 2 metals / conductors placed near to each other is known as capacitor.
→ It is an arrangement which stores charge, electric field and electrostatic potential energy.



Capacitor

$$* \quad C = \frac{+Q}{V_A - V_B}$$

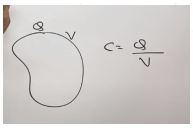


Capacitor

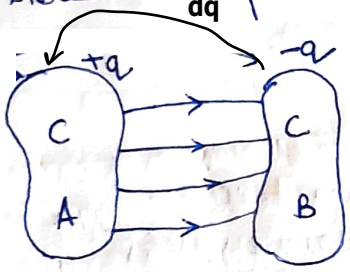
$$C = \frac{+Q}{V_A - V_B}$$

$$\Rightarrow C = \frac{+Q}{Ed}$$

$$\therefore * \boxed{C \propto \frac{1}{d}}$$



Energy stored in a capacitor:

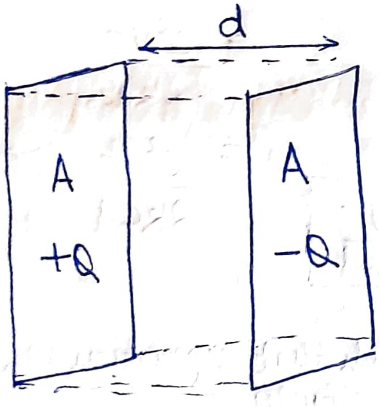


$$U = W_{ext} = \int_0^Q \frac{q}{C}\, dq$$

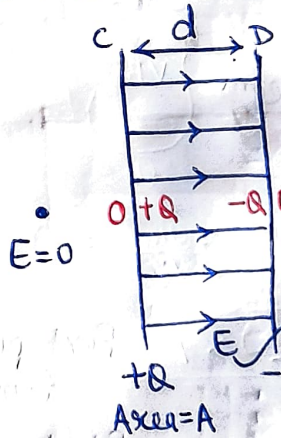
$$\{V = V_A - V_B \quad V = \frac{q}{C}\}$$

$$* \boxed{U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV} \quad (V = V_A - V_B = \text{pot. difference})$$

Parallel Plate Capacitor:



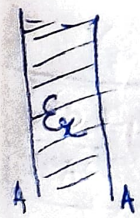
Side View:



$$d << \sqrt{A}$$

$$E = \frac{Q}{A\varepsilon_0}$$

In any other medium



$$C' = \frac{A\varepsilon}{d} = \frac{A\varepsilon_0\varepsilon_r}{d} = \varepsilon_r C$$

$$\therefore \boxed{C' = \frac{A\varepsilon_0\varepsilon_r}{d} = \varepsilon_r C}$$

$$C = \frac{+Q}{V_C - V_D} = \frac{+Q}{Ed} = \frac{Q}{\frac{Q}{A\varepsilon_0}d}$$

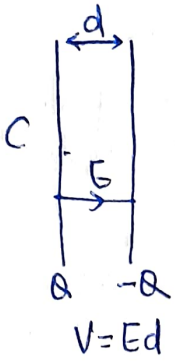
$$* \boxed{C = \frac{A\varepsilon_0}{d}} \quad (\text{in air})$$

$$* \boxed{C \propto A}$$

$$* \boxed{C \propto \frac{1}{d}}$$

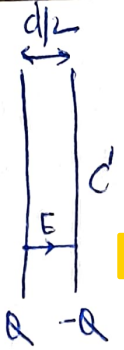
dependent on medium and shape, size, geometry.

Q



$V = Ed$

$\Rightarrow$



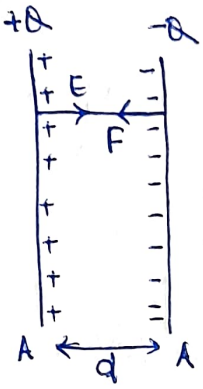
$V' = \frac{Ed}{2} = \frac{V}{2}$

Find capacitance of 2nd plate.

$C = \frac{A\varepsilon_0}{d}$

$C' = \frac{A\varepsilon_0}{d/2} = \frac{2A\varepsilon_0}{d} = 2C$

Q



Find electrostatic force b/w these plates.

$E_{net} = E = \frac{Q}{A\varepsilon_0}$

$E_1 \Rightarrow$ E.F due to $+Q$

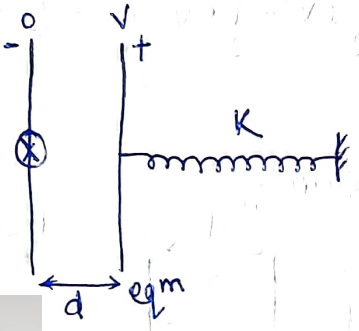
$E_1 = \frac{Q}{2A\varepsilon_0}$

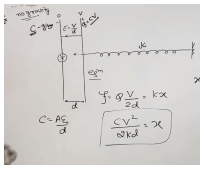
$F = QE_1 = \frac{Q^2}{2A\varepsilon_0}$

$\boxed{F = \frac{Q^2}{2A\varepsilon_0}}$

thus, if the plates of a capacitor are not hinged, then the plates will collapse into each other and neutralise each other

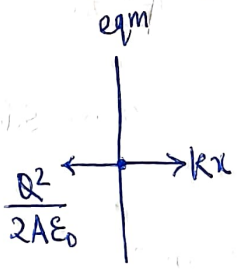
Q No gravity





Given capacitance = C

Find extension in spring.



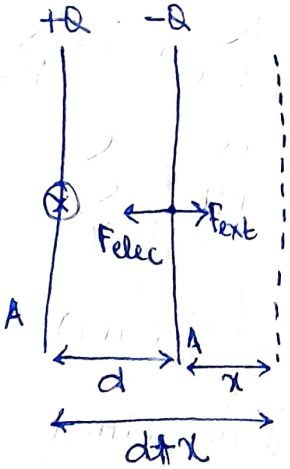
$x = \frac{Q^2}{2kA\varepsilon_0}$

$= \frac{C^2V^2}{2kA\varepsilon_0}$

$= \frac{C^2V^2}{2kCd}$ $\left\{C = \frac{A\varepsilon_0}{d}\right\}$

$\boxed{x = \frac{CV^2}{2kd}}$

Q



Fext pulls the plate such that separation increases from d to d+x slowly.

Find ① Wext ② Change in electrostatic potential energy stored in capacitor.

① $F_{ext} = F_{elec} = \frac{Q^2}{2A\varepsilon_0}$

charges will remain same on pulling out

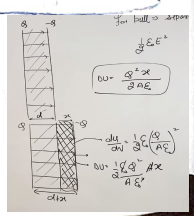
$W_{ext} = F_{ext}x = \frac{Q^2x}{2A\varepsilon_0}$

$\therefore \boxed{W_{ext} = \frac{Q^2x}{2A\varepsilon_0}}$

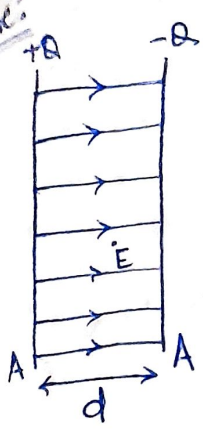
② $U_i = \frac{Q^2}{2C} = \frac{Q^2d}{2A\varepsilon_0}$

$U_f = \frac{Q^2}{2C} = \frac{Q^2(d+x)}{2A\varepsilon_0}$

$\boxed{\Delta U = \frac{Q^2x}{2A\varepsilon_0}}$



Note:



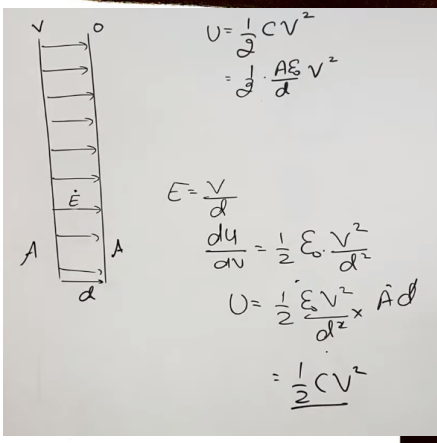
$$U = \frac{Q^2}{2C} = \frac{Q^2 d}{2A\varepsilon_0}$$

Another way to find this:

$$E = \frac{Q}{A\varepsilon_0}$$

$$\frac{dU}{dV} = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \frac{Q^2}{A^2\varepsilon_0^2} = \frac{Q^2}{2A^2\varepsilon_0}$$

$$U = \frac{Q^2}{2A^2\varepsilon_0} \times Ad = \frac{Q^2 d}{2A\varepsilon_0} = \frac{Q^2}{2C}$$



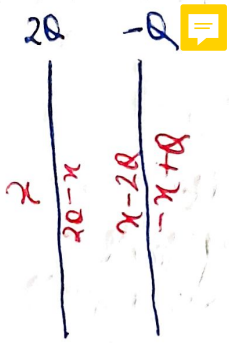
$$U = \frac{1}{2}CV^2 = \frac{1}{2}\cdot\frac{A\varepsilon}{d}V^2$$

$$E = \frac{V}{d}$$

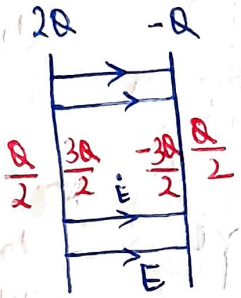
$$\frac{du}{dv} = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2}$$

$$U = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2} \times Ad = \frac{1}{2}CV^2$$

Q

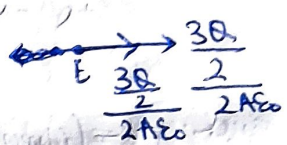


Given capacitance = C. Find potential difference b/w plates.
Let area of plates be A and dist b/w them be d.



$$x = -x + Q \Rightarrow \boxed{x = Q/2}$$

$$E = \frac{Q}{2A\varepsilon_0}$$



$$E = \frac{3Q}{2} \times \frac{2}{2A\varepsilon_0} = \frac{3Q}{2A\varepsilon_0}$$

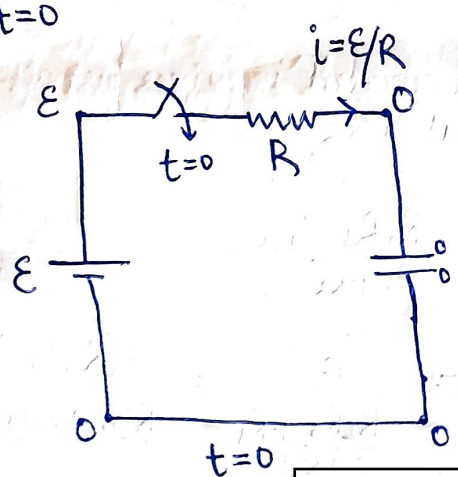
$$\therefore \boxed{\Delta V = \frac{3Q}{2C}}$$

$$\Delta V = Ed = \frac{3Qd}{2A\varepsilon_0} = \frac{3Q}{2C}$$

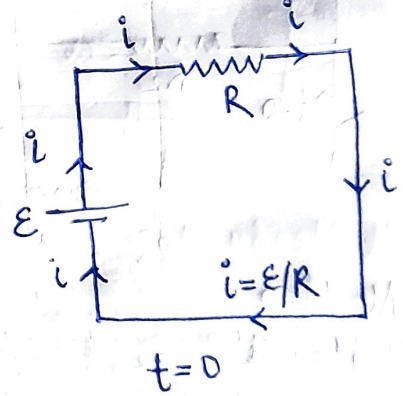
Circuit Solution of R-C circuit (for $t=0$ and $t \to \infty$):

steady state

(i) At $t=0$

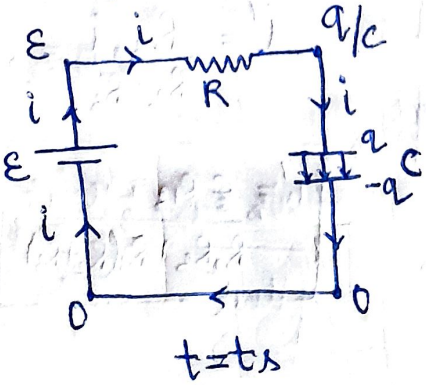


C {initially uncharged} $\Rightarrow Q=0$



* At $t=0$, if capacitor is uncharged, it will behave like short wire.

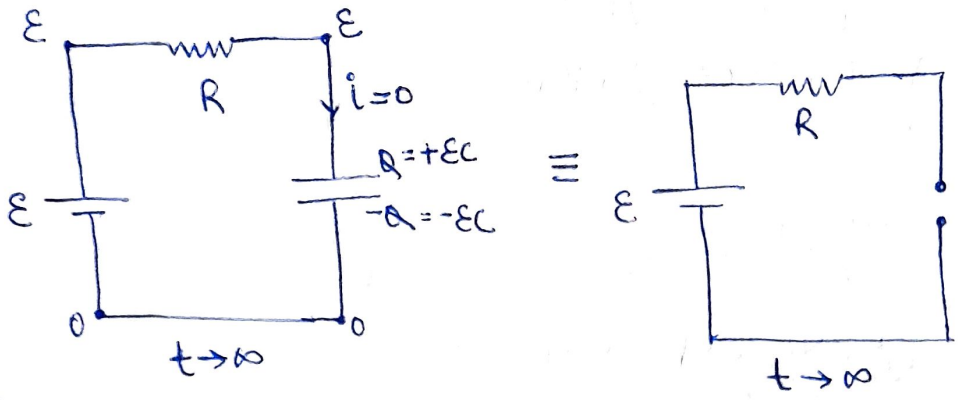
Now, at time t.



$$i = \frac{\varepsilon - \frac{q}{C}}{R}$$

As $t \uparrow$, $q \uparrow$, $i \downarrow$

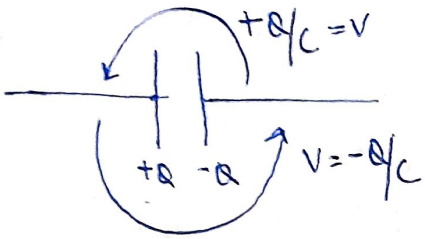
(ii) At steady state ($t \to \infty$):



$Q = CV = C(\varepsilon - 0) = \varepsilon C$

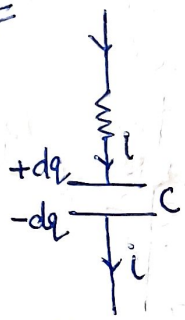
★ At steady state, capacitor is behaving like a broken wire.

#



Trick: $V = -\dfrac{(\text{Charge on first plate})}{C}$

#



After time dt

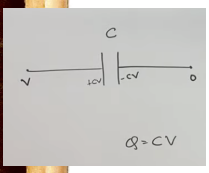
$dq = i\,dt$

$dt \to$ very small

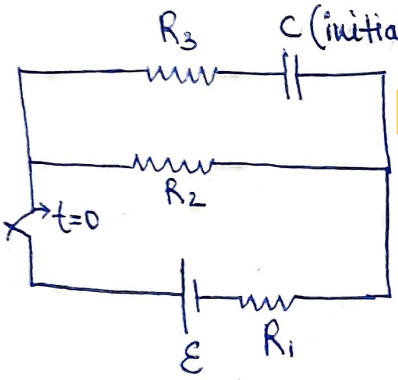
$\Rightarrow dq \to$ very small

★ Charge on a capacitor doesn't change in infinitely small interval.

★ Charge on a capacitor changes gradually with time



R



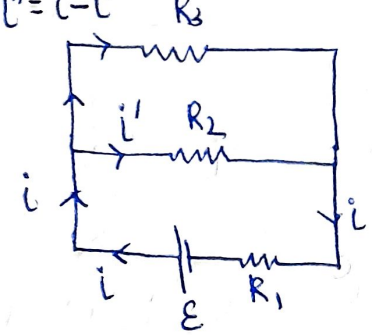
① Find current in wires just after switch is on.

$$R_{eq} = \frac{R_2R_3}{R_2+R_3} + R_1$$

$$i = \frac{\varepsilon}{R_{eq}} = \frac{\varepsilon}{\frac{R_2R_3}{R_2+R_3}+R_1} = \frac{\varepsilon(R_2+R_3)}{R_2R_3+R_1(R_2+R_3)}$$

At $t=0$

$i'' = i - i'$

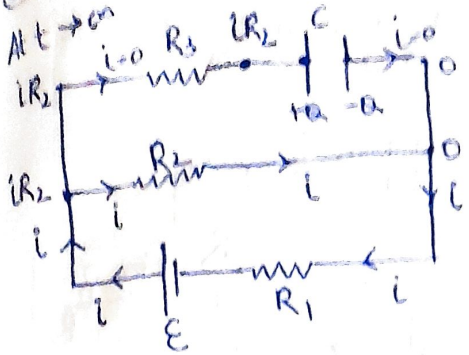


$$i' = \frac{R_3\,i}{R_2+R_3} = \frac{R_3}{R_2+R_3} \times \frac{\varepsilon(R_2+R_3)}{R_2R_3+R_1(R_2+R_3)}$$

$$\boxed{i' = \frac{\varepsilon R_3}{R_2R_3+R_1(R_2+R_3)}}$$

$$\boxed{i'' = \frac{\varepsilon R_2}{R_2R_3+R_1(R_2+R_3)}}$$

Ex Find current in each wire, and charge on capacitor at steady state.

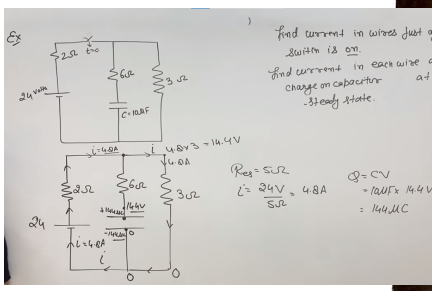


$$V = iR_2 = \frac{\varepsilon R_2}{R_1+R_2}$$

$$Q = CV = \frac{C\varepsilon R_2}{R_1+R_2}$$

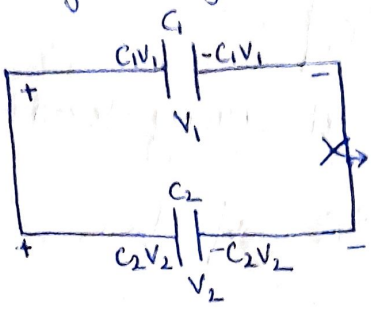
$$\therefore \boxed{Q = \frac{C\varepsilon R_2}{R_1+R_2}}$$

$$i = \frac{\varepsilon}{R_1+R_2}$$



Sharing of charge between 2 capacitors:

Ex

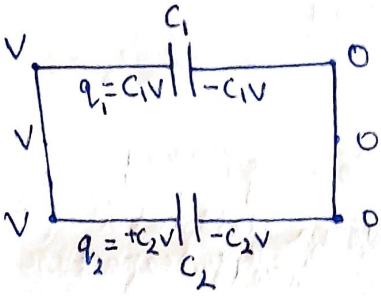


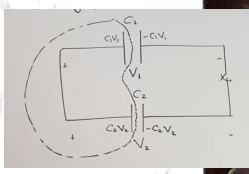
① Find final p.d across each capacitor
② Find final charge on each capacitor
③ Find heat dissipated.

① Using charge conservation:

$$C_1V_1 + C_2V_2 = C_1V + C_2V$$

$$\boxed{V = \frac{C_1V_1 + C_2V_2}{C_1+C_2}}$$





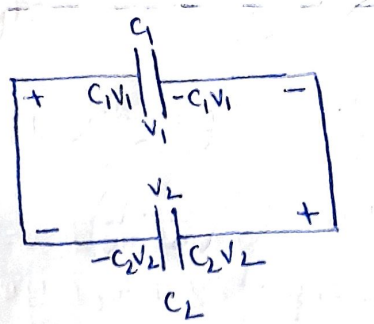
② $\boxed{q_1 = C_1V = \frac{C_1(C_1V_1+C_2V_2)}{C_1+C_2}}$ $\boxed{q_2 = \frac{C_2(C_1V_1+C_2V_2)}{C_1+C_2}}$

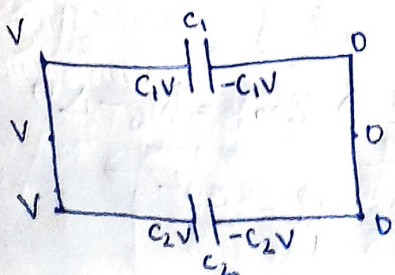
③ $\text{Heat} = U_i - U_f = \left\{\frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2\right\} - \left\{\frac{1}{2}C_1V^2 + \frac{1}{2}C_2V^2\right\}$

$$\boxed{\text{Heat} = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1-V_2)^2}$$

Finally, current in circuit is zero.

Ex

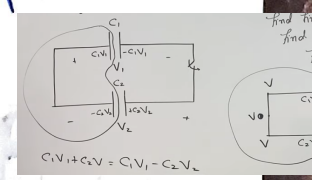




① Find final p.d across each capacitor

c/c → $C_1V_1 - C_2V_2 = C_1V + C_2V$

$$\boxed{V = \frac{C_1V_1 - C_2V_2}{C_1+C_2}}$$



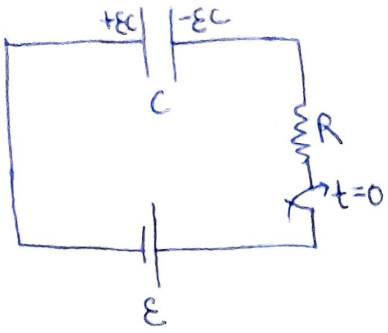
③ $\text{Heat} = U_i - U_f = \left\{\frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2\right\} - \left\{\frac{1}{2}C_1V^2 + \frac{1}{2}C_2V^2\right\}$

$$\boxed{\text{Heat} = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1+V_2)^2}$$

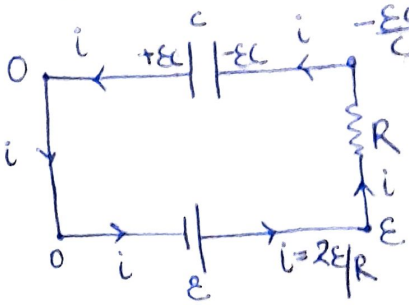
all these results could also be obtained by replacing V2 by -V2

Q

Initially, a capacitor is charged with a battery of EMF E. After charging, the polarity of the battery is reversed

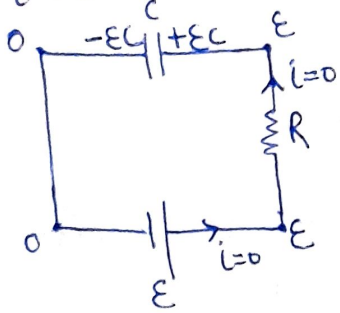


① Find current in circuit just after switch is on.



$$i = \frac{2\varepsilon}{R}$$

At $t \to \infty$



② Find work done by battery.

$$W_b = \varepsilon(Q_{given}) = \varepsilon(\varepsilon C - (-\varepsilon C)) = \varepsilon(2\varepsilon C)$$

$$W_b = 2\varepsilon^2 C$$

③ Find heat lost in process

$$W_b = \Delta U + \text{Heat}$$

$$\text{Heat} = W_b - \Delta U = 2\varepsilon^2 C$$

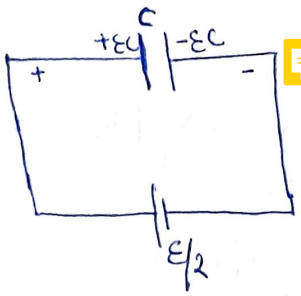
$$\text{Heat} = 2\varepsilon^2 C$$

④ Find change in electric energy stored in capacitor.

$$U_i = \frac{1}{2}C\varepsilon^2 \qquad U_f = \frac{1}{2}C\varepsilon^2$$

$$\Delta U = 0$$

Q

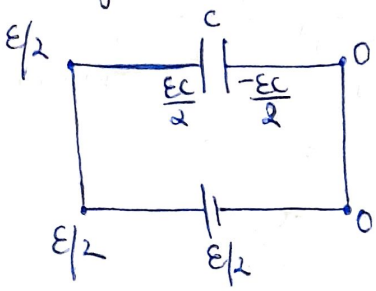


Find ① Work done by battery

② change in electrostatic potential energy stored in capacitor.

③ Heat lost.

At steady state:



① $W_b = \varepsilon Q = \frac{\varepsilon}{2}\left(\frac{\varepsilon C}{2} - \varepsilon C\right) = -\frac{\varepsilon^2 C}{4}$

② $U_i = \frac{1}{2}C\varepsilon^2 \qquad U_f = \frac{1}{2}C\left(\frac{\varepsilon}{2}\right)^2 = \frac{C\varepsilon^2}{8}$

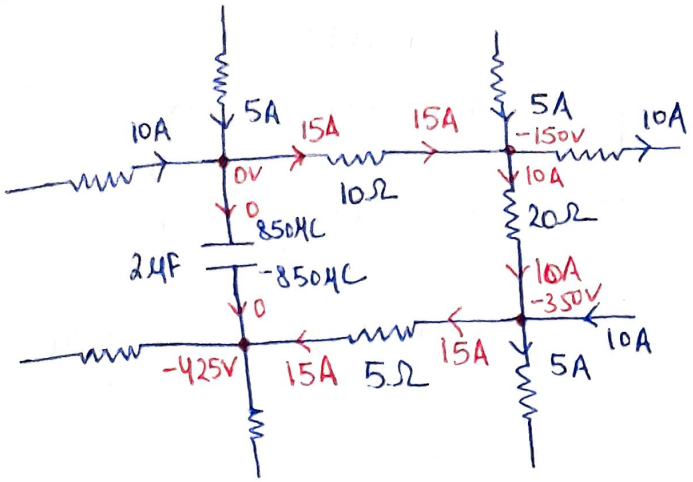
$$\Delta U = U_f - U_i = C\varepsilon^2\left(\frac{1}{8} - \frac{1}{2}\right)$$

$$\Delta U = -\frac{3C\varepsilon^2}{8}$$

③ $W_b = \Delta U + \text{Heat} \Rightarrow \text{Heat} = W_b - \Delta U$

$$\text{Heat} = C\varepsilon^2\left\{\frac{3}{8} - \frac{1}{4}\right\} = \frac{C\varepsilon^2}{8}$$

Q



Capacitor is in steady state.

② Also find electrostatic potential energy stored in capacitor.

① Find charge on capacitor.

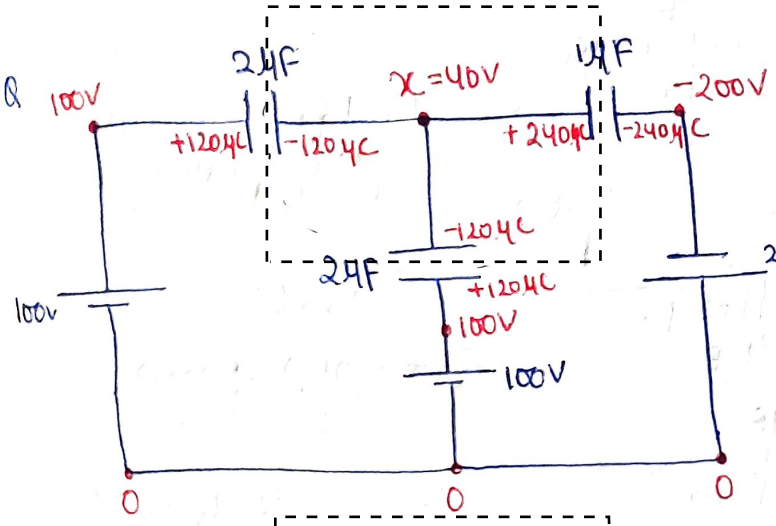
① $Q = CV = 2\mu F \times 425V$

$= 850\mu C$

② $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 2 \times (425)^2 \mu J$

$U = (425)^2 \mu J$

Q



Initially all capacitor are uncharged. Find final charge on each capacitor.

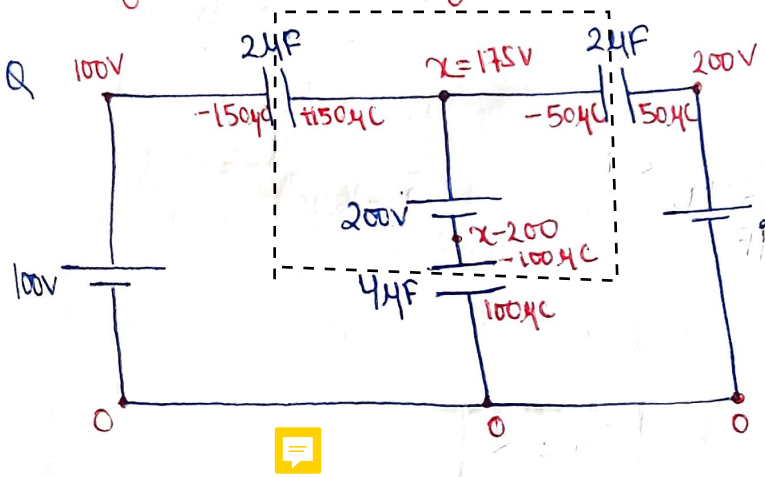
Using c/c for x:

$0+0+0 = 2(x-100) + 1(x+200) + 2(x-100)$

$\Rightarrow 0 = 2x-200+x+200+2x-200$

$= 5x = 200 \Rightarrow \boxed{x = 40V}$

Q



Initially all capacitors are uncharged. Find final charge on each capacitor.

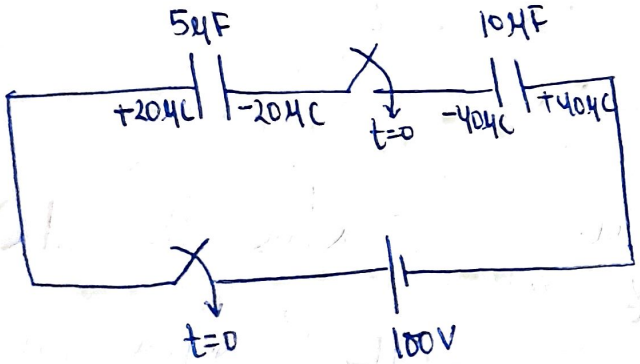
Using charge conservation for x:

$0+0+0 = 2(x-100) + 2(x-200) + 4(x-200)$

$\Rightarrow 0 = 2x-200+2x-400+4x-800$

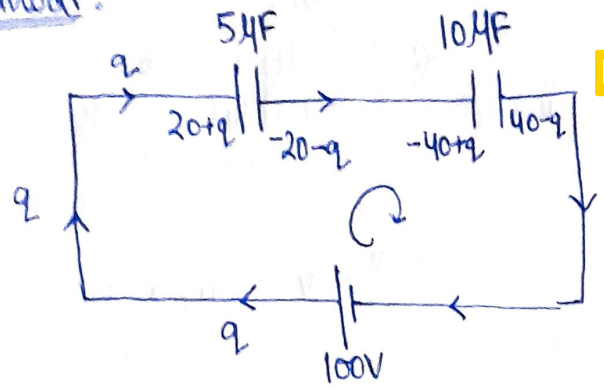
$\boxed{x = 175V}$

Q



Find final charge on each capacitor.

Method1:



Using KVL:

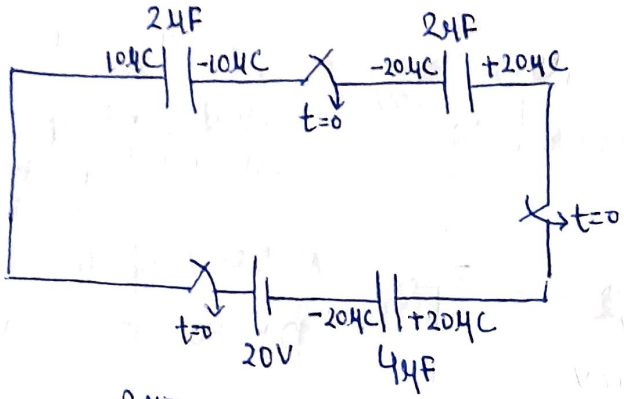
$$100-\frac{(20+q)}{5}-\frac{(-40+q)}{5}=0$$

$$100-4-\frac{q}{5}+4-\frac{q}{10}=0$$

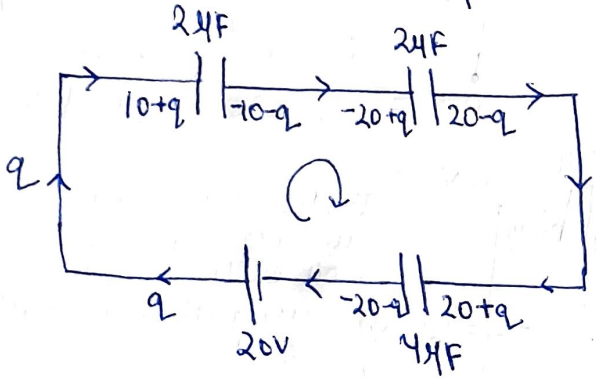
$$3q=1000 \Rightarrow \boxed{q=\frac{1000}{3}\mu C}$$

Method2: Use KCL.

Q



Find final charge on each capacitor.



Using KVL:

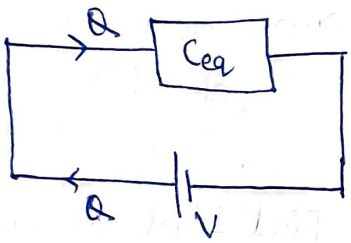
$$20-\frac{(10+q)}{2}-\frac{(-20+q)}{2}-\frac{(20+q)}{4}=0$$

$$=20-5-\frac{q}{2}+10-\frac{q}{2}-5-\frac{q}{4}=0$$

$$20=\frac{5q}{4}$$

$$\boxed{q=16\mu C}$$

Equivalent Capacitance:



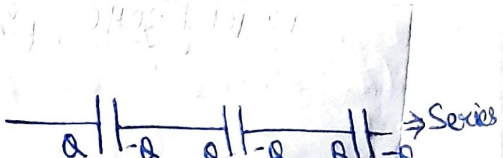
$$\boxed{C_{eq}=\frac{Q}{V}}$$

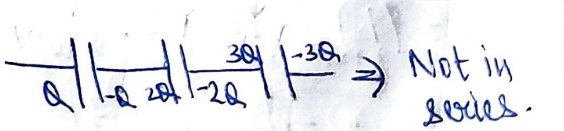
equivalent capacitance of a system of capacitors is the capacitance which consumes the same amount of charge from the battery as consumed by the system of capacitors from the same battery and stores the same amount of energy as stored by the system of capacitors

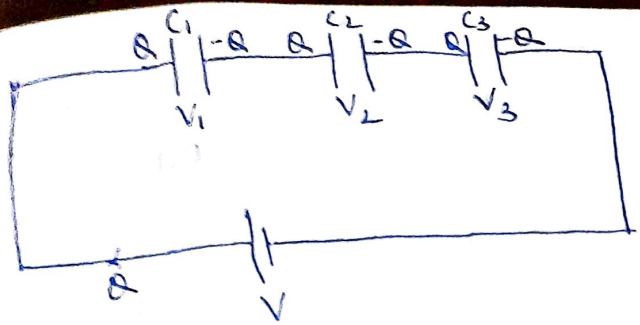
Series-Combination:

→ They must be connected end to end.

→ Charge on each capacitor must be same







Capacitors are initially uncharged

Using KVL:

$$V - \frac{Q}{C_1} - \frac{Q}{C_2} - \frac{Q}{C_3} = 0$$

$$\Rightarrow \frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\Rightarrow \boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

$V_1 = Q/C_1$
$V_2 = Q/C_2$
$V_3 = Q/C_3$

$$\boxed{V_1 : V_2 : V_3 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C_3}}$$

$$\boxed{V \propto \frac{1}{C}}$$

Energy Stored $\Rightarrow U_1 = \frac{Q^2}{2C_1}$ $U_2 = \frac{Q^2}{2C_2}$ $U_3 = \frac{Q^2}{2C_3}$

$$\boxed{U_{net} = U_1 + U_2 + U_3 = \frac{Q^2}{2C_{eq}}}$$

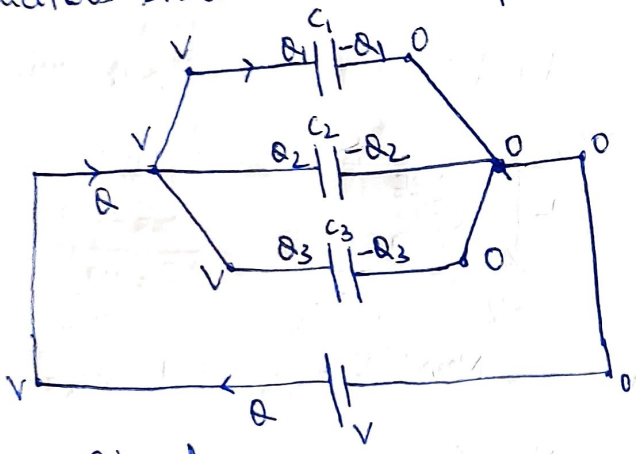
$$\boxed{U \propto \frac{1}{C}}$$

$$\boxed{U_1 : U_2 : U_3 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C_3}}$$

Special case:
For two capacitors in series:-

$$\boxed{C_{eq} = \frac{C_1 C_2}{C_1 + C_2}}$$

Parallel Combination:
→ Capacitors should have same potential difference.



$Q_1 = C_1 V$
$Q_2 = C_2 V$
$Q_3 = C_3 V$

$$\boxed{Q_1 : Q_2 : Q_3 = C_1 : C_2 : C_3}$$

$$\Rightarrow Q = Q_1 + Q_2 + Q_3 = (C_1 + C_2 + C_3)V$$

$$\Rightarrow \frac{Q}{V} = (C_1 + C_2 + C_3)$$

$$\boxed{C_{eq} = (C_1 + C_2 + C_3)}$$

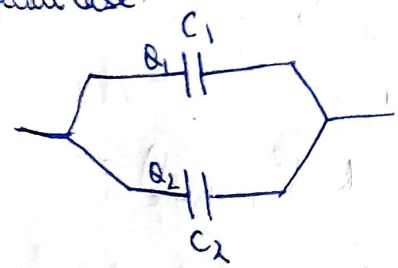
Energy Stored:

$U_1 = \frac{1}{2} C_1 V^2$ $U_2 = \frac{1}{2} C_2 V^2$ $U_3 = \frac{1}{2} C_3 V^2$

$$\boxed{U_{net} = \frac{1}{2} C_{eq} V^2}$$

$$\boxed{U_1 : U_2 : U_3 = C_1 : C_2 : C_3}$$

Special case:



$$Q_1 + Q_2 = Q \quad (1)$$

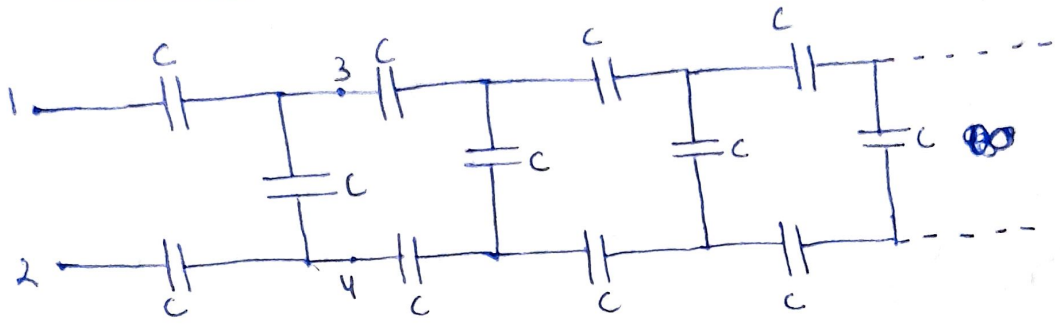
$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$\rightarrow \boxed{\frac{Q_1}{Q_2} = \frac{C_1}{C_2}} \quad (2)$$

$$\boxed{Q_1 = \frac{C_1}{C_1 + C_2} Q}$$

$$\boxed{Q_2 = \frac{C_2}{C_1 + C_2} Q}$$

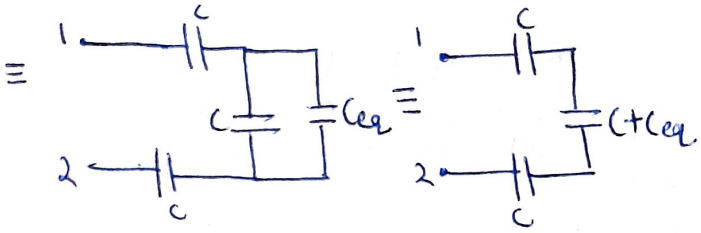
Q



Find $(C_{eq})_{12}$ a

$C_{12} = C_{eq} = C_{34}$

$\equiv$



$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C+C_{eq}}$$

$$= \frac{1}{C_{eq}} = \frac{2}{C} + \frac{1}{C+C_{eq}}$$

$$= \frac{1}{C_{eq}} = \frac{2C+2C_{eq}+C}{C(C+C_{eq})}$$

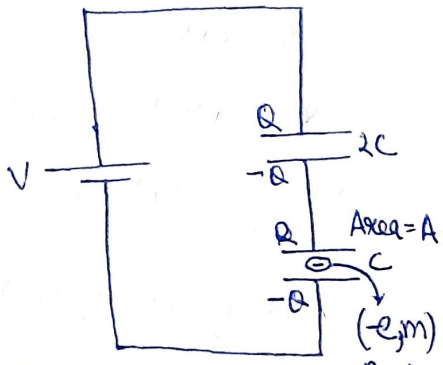
$$= C^2 + CC_{eq} = 3CC_{eq} + 2C_{eq}^2$$

$$\Rightarrow 2C_{eq}^2 + 2CC_{eq} - C^2 = 0$$

$$\Rightarrow C_{eq} = \frac{-2C + \sqrt{4C^2 + 4\times 2C^2}}{4}$$

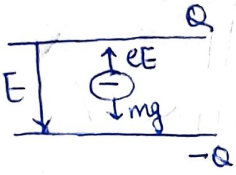
$$\boxed{C_{eq} = \frac{-2C + \sqrt{12}C}{4}}$$

Q Vertically



$$\therefore \boxed{V = \frac{3mgA\varepsilon_0}{2Ce}}$$

Find V such that $e^{\ominus}$ is at equilibrium



For equilibrium:

$eE = mg$

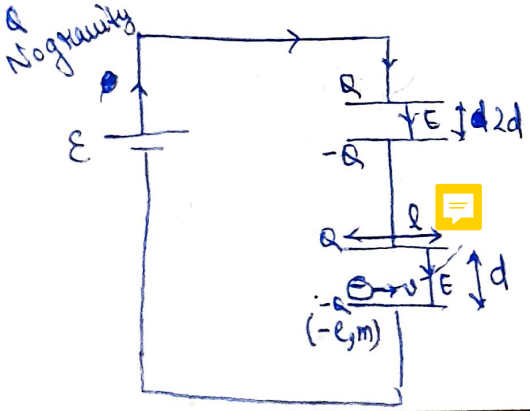
$$\boxed{E = \frac{mg}{e}}$$

Now, $E = \frac{Q}{A\varepsilon_0} = \frac{mg}{e} \Rightarrow \boxed{Q = \frac{mgA\varepsilon_0}{e}}$

Now, $C_{eq} = \frac{2C\cdot C}{2C+C} = \frac{2C}{3}$

$\Rightarrow Q = C_{eq}V \Rightarrow V = \frac{Q}{C_{eq}} = \frac{3mgA\varepsilon_0}{2Ce}$

Q No gravity

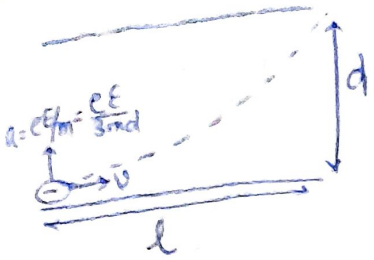


Find min^m value of V such that electron doesn't strike the plate.

Using KVL:

$$\varepsilon - Ed(2d) - E(d) = 0$$

$$\boxed{E = \frac{\varepsilon}{3d}}$$



$$S_y = u_y t + \frac{1}{2} a_y t^2$$

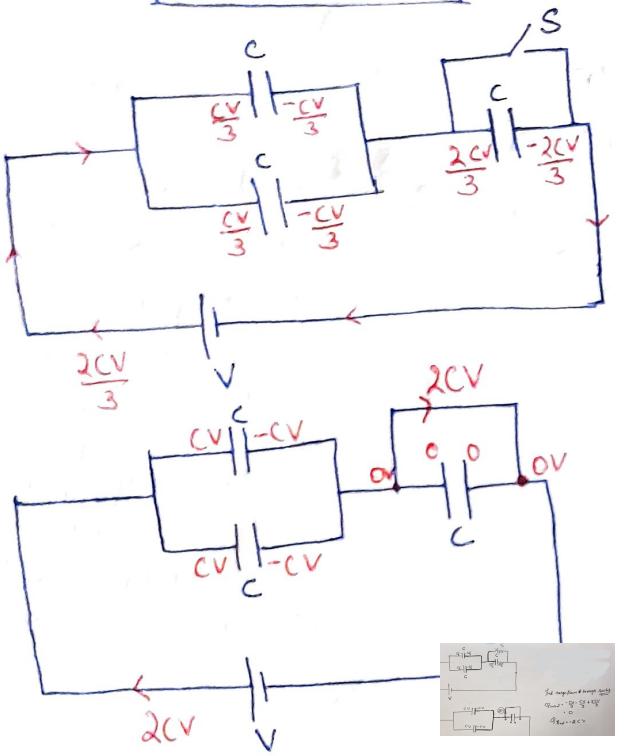
$$\Rightarrow d = 0 + \frac{1}{2}\left(\frac{eE}{3md}\right) t^2 \Rightarrow \boxed{t = d\sqrt{\frac{6m}{eE}}}$$

$$S_x > l$$

$$\Rightarrow vt > l \Rightarrow v > \frac{l}{t} \Rightarrow \boxed{v > \frac{l}{d}\sqrt{\frac{eE}{6m}}}$$

$$\boxed{v_{min} = \frac{l}{d}\sqrt{\frac{eE}{6m}}}$$

Q



① Find charge on each capacitor when switch is off.

$$C_{eq} = \frac{2C \cdot C}{2C + C} = \frac{2C}{3}$$

$$\boxed{Q = \frac{2CV}{3}}$$

② Find charge on each capacitor after switch is on.

$$C_{eq} = 2C$$

$$Q = 2CV$$

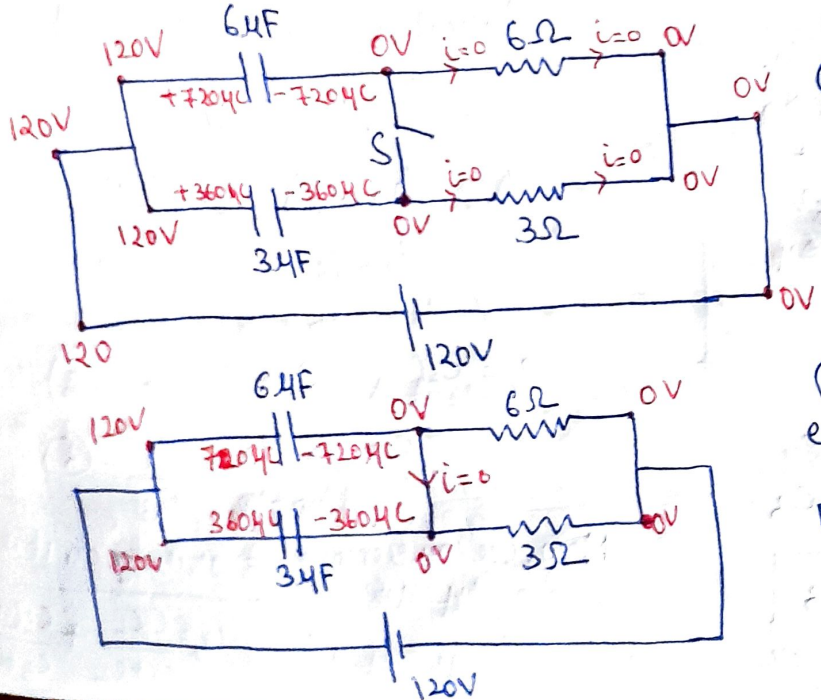
③ Find charge flown through switch.

$$Q_{initial} = -\frac{CV}{3} + \frac{-CV}{3} + \frac{2CV}{3} = 0$$

$$Q_{final} = -CV + -CV + 0 = -2CV$$

$\therefore$ 2CV charge flows from switch.

Q



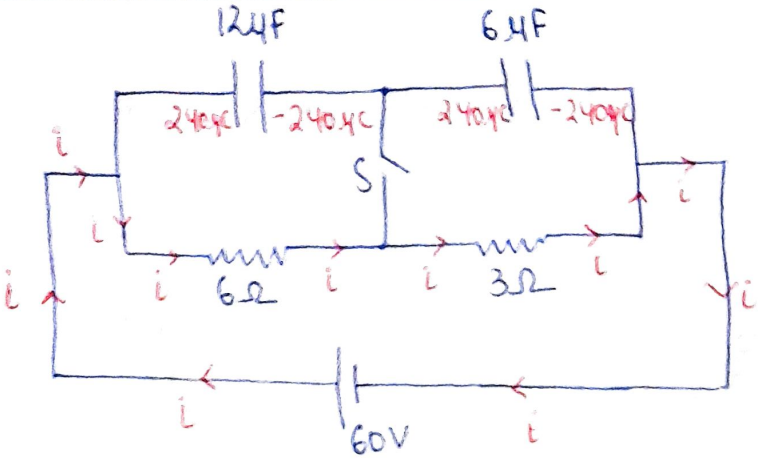
① Find charge on each capacitor when switch is open.

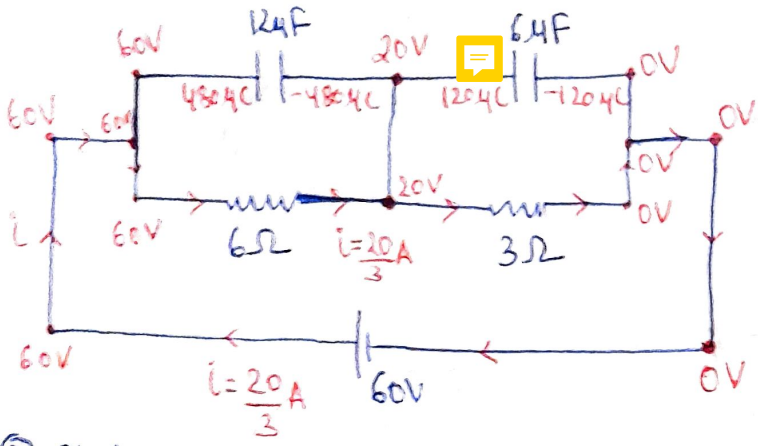
Current in wire is zero at steady state.

② Final final charge on each capacitor after switch is on.

Even after switch is on, at steady state, no current flows as $\Delta V = 0$ across S. Charge remains same.

Q





① Find charge on each capacitor when switch is open.

At steady state, no current flows through capacitors.

$i = \frac{60}{6+3} = \frac{20}{3}$ A

$C_{eq} = \frac{12 \times 6}{18} = 4\mu F$

$Q = C_{eq} V = 4 \times 60 = 240\mu C$

② Find final charge on each capacitor after switch is on.

When switch is closed, some current and charge flows through switch for some time until steady state is attained. Now current through capacitors & switch is again zero.

Charge on $12\mu F = 12 \times 40 = 480\mu C$

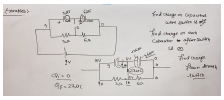
Charge on $6\mu F = 6 \times 20 = 120\mu C$

③ Find charge flown through switch.

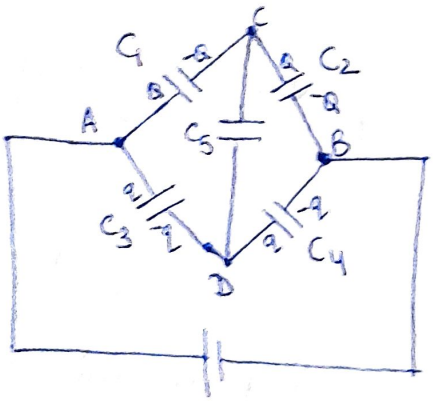
$Q_{initial} = -240 + 240 = 0$

$Q_{final} = -480 + 120 = -360$

∴ Charge flown through $S = 360\mu C$



Wheatstone Bridge:



If charge on C_5 is zero, we say wheatstone bridge is balanced.

$\Rightarrow \boxed{V_C = V_D}$

$Q = C_1(V_A - V_C) = C_2(V_C - V_B)$

$\frac{V_A - V_C}{V_C - V_B} = \frac{C_2}{C_1}$ —— (1)

$q = C_3(V_A - V_D) = C_4(V_D - V_B)$

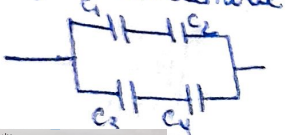
$\frac{V_A - V_D}{V_D - V_B} = \frac{C_4}{C_3}$ —— (2)

As $V_C = V_D \Rightarrow \frac{C_2}{C_1} = \frac{C_4}{C_3}$

$\therefore \boxed{C_1 C_4 = C_2 C_3}$

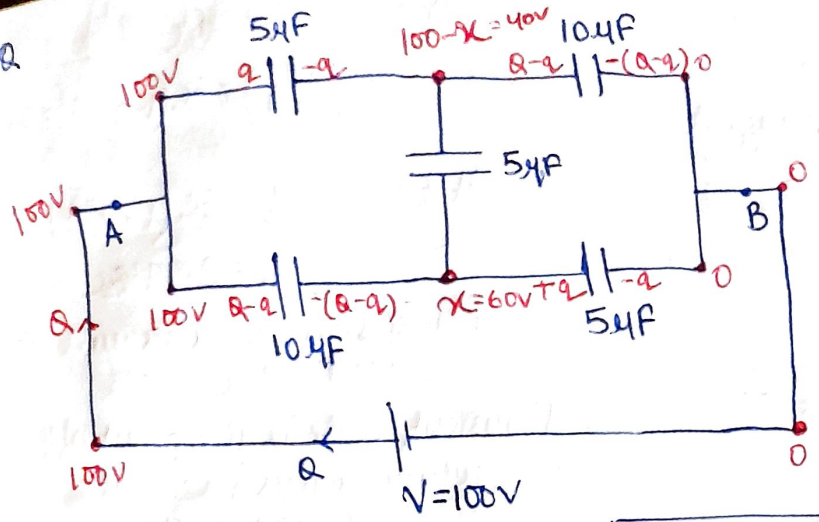
↳ Condition for balanced wheatstone bridge

Now, we can remove C_5, as charge on it is zero



$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4}$

Q



$C_{eq} = \frac{Q}{100} = \frac{700}{100} = 7\mu F$

$\therefore \boxed{C_{eq} = 7\mu F}$

Find $(C_{eq})_{AB}$

As $C_1C_4 \neq C_2C_3$

$\Rightarrow$ WSB is not balanced.

Using c/c at 100-x junction:

$5(-x) + 10(100-x) + 5(100-2x) = 0$

$\Rightarrow -x + 200 - 2x + 100 - 2x = 0$

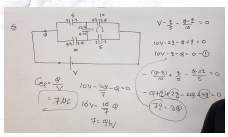
$5x = 300$

$\boxed{x = 60V}$

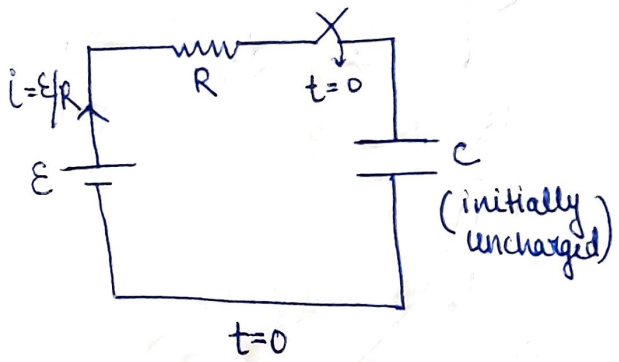
$q = 5 \times 60 = 300\mu C \Rightarrow$

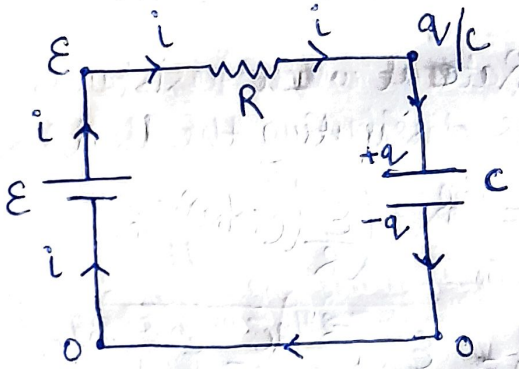
$Q - q = 40 \times 10 = 400$

$\boxed{Q = 700\mu C}$



Charging of a capacitor:





At any time t:

Using KVL $\Rightarrow \varepsilon - iR - \frac{q}{C} = 0$ — (1)

$i = \frac{dq}{dt}$ — (2)

From (1) and (2)

(I) $\varepsilon C - q = iRC$

$= i = \frac{\varepsilon C - q}{RC} = \frac{dq}{dt}$

$= \int_0^q \frac{dq}{\varepsilon C - q} = \int_0^t \frac{dt}{RC}$

$= -\ln(\varepsilon C - q)_0^q = \frac{t}{RC}$

$= \ln(\varepsilon C - q) - \ln(\varepsilon C) = \frac{-t}{RC}$

$= \ln\left(\frac{\varepsilon C - q}{\varepsilon C}\right) = \frac{-t}{RC}$

$= \boxed{1 - \frac{q}{\varepsilon C} = e^{-t/RC}}$

$\therefore \boxed{q = \varepsilon C\left(1 - e^{-t/RC}\right)}$ *

Now $RC = \tau$ = Time constant.

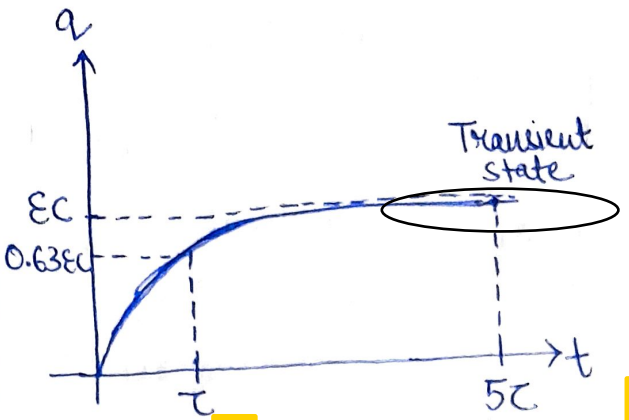
$\boxed{q = \varepsilon C\left(1 - e^{-t/\tau}\right)}$ *

At $t = 0 \Rightarrow q = 0$

At $t \to \infty \Rightarrow q = \varepsilon C\left(1 - e^{-\infty}\right)$

$= \varepsilon C\left(1 - \frac{1}{e^{\infty}}\right)$

$\boxed{q = \varepsilon C}$



(II) Now, $q = \varepsilon C(1 - e^{-t/RC})$

$$i = \frac{dq}{dt} = \varepsilon C\left(0 - \frac{e^{-t/RC}}{-RC}\right)$$

$$\star \boxed{i = \frac{\varepsilon}{R} e^{-t/RC} = \frac{\varepsilon}{R} e^{-t/\tau}}$$

(III) Rate at which resistance is dissipating the heat:

$$P_R = i^2 R = \left(\frac{\varepsilon}{R}(e^{-t/RC})\right)^2 R$$

$$\star \boxed{P_R = \frac{\varepsilon^2}{R} e^{-2t/RC} = \frac{\varepsilon^2}{R} e^{-2t/\tau}}$$

(IV) Heat lost By resistance:

$$= \int_0^\infty P_R \, dt = \frac{\varepsilon^2}{R} \int_0^\infty e^{-2t/RC} \, dt$$

$$= \frac{\varepsilon^2}{R}\left(\frac{RC}{-2}\right)\left\{e^{-2t/RC}\right\}_0^\infty$$

$$= -\frac{\varepsilon^2 C}{2}\{0 - 1\}$$

$$\star \boxed{\text{Heat} = \frac{1}{2} C\varepsilon^2}$$

(V) At $t = \tau$, $q = \varepsilon C(1 - e^{-1})$

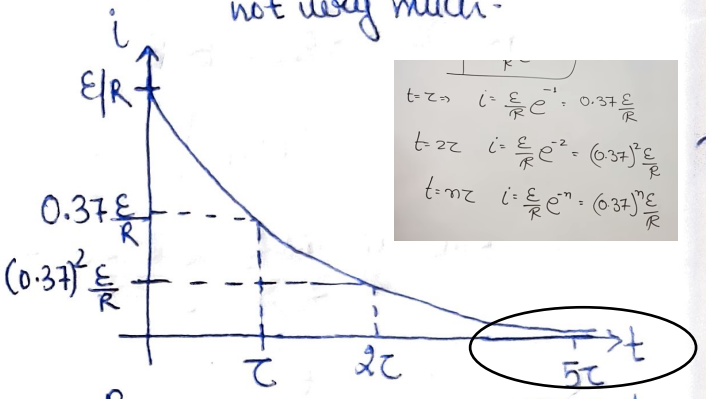
$$= \varepsilon C\left(1 - \frac{1}{2.7}\right) = \varepsilon C(1 - 0.37) = 0.63\varepsilon C$$

At $t = 5\tau$, $q = \varepsilon C(1 - e^{-5})$

$$q = 0.993\varepsilon C \approx \varepsilon C$$

∴ Practically, capacitor will get charged after 5τ.

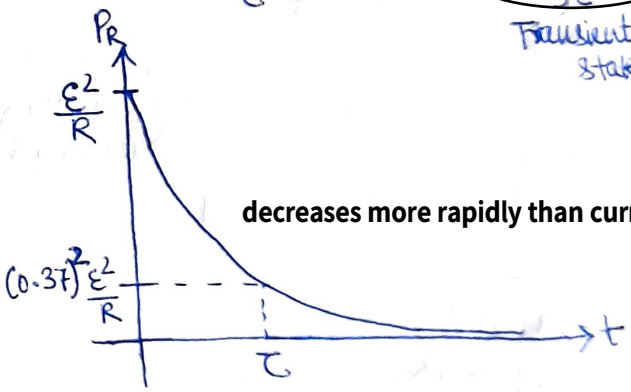
Transient state is the point after ★ which changes in values are not very much.



$t = \tau \Rightarrow i = \frac{\varepsilon}{R} e^{-1} = 0.37\frac{\varepsilon}{R}$

$t = 2\tau \quad i = \frac{\varepsilon}{R} e^{-2} = (0.37)^2 \frac{\varepsilon}{R}$

$t = n\tau \quad i = \frac{\varepsilon}{R} e^{-n} = (0.37)^n \frac{\varepsilon}{R}$



(VI) decreases more rapidly than current

★ Area under P_R V/s t graph gives heat lost.

Alternate:

$W_b = \Delta U + \text{Heat}$

$\text{Heat} = W_b - \Delta U$

$\text{Heat} = \varepsilon(\varepsilon C) - \frac{1}{2} C\varepsilon^2$

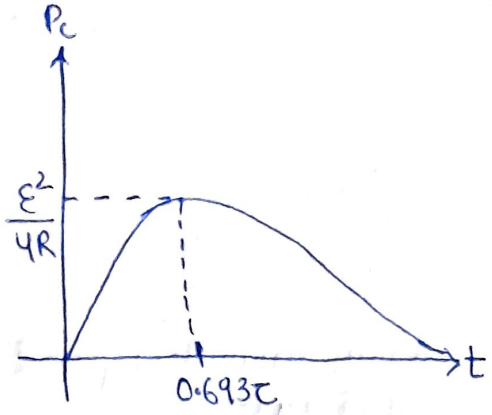
$$\boxed{\text{Heat} = \frac{1}{2} C\varepsilon^2}$$

(v) Rate at which capacitor is storing Electric Energy:

$$P_c = V_c i = \frac{q}{C} i$$

$$P_c = \frac{\varepsilon R}{R}\left(1-e^{-t/RC}\right)\frac{\varepsilon}{R}\left(e^{-t/RC}\right)$$

$$\boxed{P_c = \frac{\varepsilon^2}{R}\left(e^{-t/RC} - e^{-2t/RC}\right) = \frac{\varepsilon^2}{R}\left(e^{-t/\tau} - e^{-2t/\tau}\right)}$$



At $t=0$, $q=0$, $P_c=0$
At $t\to\infty$, $i=0$, $P_c=0$

$$P_{c\,max} \Rightarrow \frac{dP_c}{dt} = \frac{\varepsilon^2}{R}\left(\frac{-1}{\tau}e^{-t/\tau} - \left(\frac{-2}{\tau}\right)e^{-2t/\tau}\right) = 0$$

$$= e^{-t/\tau} = 2e^{-2t/\tau}$$

$$= e^{t/\tau} = 2$$

$$= \frac{t}{\tau} = \ln 2 \Rightarrow \boxed{t = \tau \ln 2 = 0.693\tau}$$

Max value of P_c

$$P_c = \frac{\varepsilon^2}{R}\left(e^{-t/\tau} - e^{-2t/\tau}\right)$$

$$= \frac{\varepsilon^2}{R}\left(e^{-\ln 2} - e^{-2\ln 2}\right)$$

$$= \frac{\varepsilon^2}{R}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{\varepsilon^2}{4R}$$

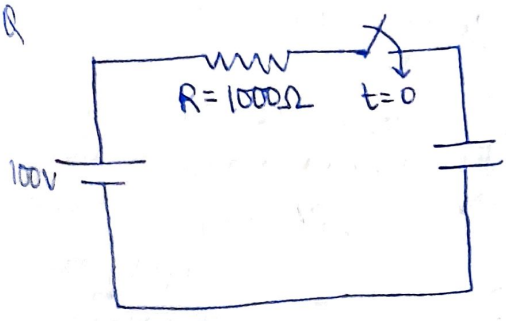
(vi) Area under P_c-t graph:

$$\text{Area} = \int_0^\infty P_c\,dt = \frac{\varepsilon^2}{R}\int_0^\infty \left(e^{-t/RC} - e^{-2t/RC}\right)dt = \frac{\varepsilon^2}{R}\left(-RC\,e^{-t/RC} + \frac{RC}{2}e^{-2t/RC}\right)_0^\infty$$

$$= \varepsilon^2 C\left((0-0) - \left(-1+\tfrac{1}{2}\right)\right) = \frac{1}{2}C\varepsilon^2$$

∴ Area under P_c-t graph gives energy stored by capacitor.

Q



$\tau = RC = 1000\,\Omega \times 100\,\mu F$
$= 10^5\,\mu Sec$
$\boxed{\tau = 0.1\,s}$

At any time t, find ① q ② i ③ V_R ④ V_c

① $q = \varepsilon C\left(1-e^{-t/\tau}\right) = 10^2 \times 10^{-4}\left(1-e^{-t/0.1}\right)$

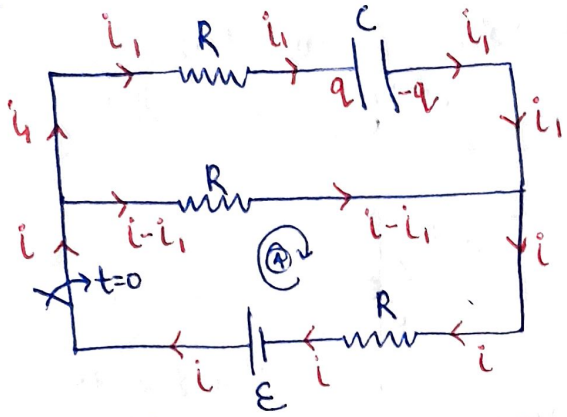
$\boxed{q = 0.01\left(1-e^{-10t}\right)\text{ Coulombs}}$

② $i = \frac{dq}{dt} = 0.01\left(0-(-10)e^{-10t}\right) = 0.1\,e^{-10t}$ A

③ $V_R = iR = 0.1\times10^3\,e^{-10t} = 100\,e^{-10t}$ Volts

④ $V_c = \frac{q}{C} = \frac{0.01}{10^{-4}}\left(1-e^{-10t}\right) = 100\left(1-e^{-10t}\right)$ Volts

Q**



Find charge on capacitor at any time t.

$$i_1 = \frac{dq}{dt} \quad -(1)$$

Using KVL:

In complete circuit:

$$\varepsilon - i_1R - \frac{q}{C} - iR = 0 \quad -(2)$$

In loop (A):

$$\varepsilon - (i-i_1)R - iR = 0 \quad -(3)$$

$$= \varepsilon - 2iR + i_1R = 0$$

$$= iR = \frac{\varepsilon + i_1R}{2} \quad -(3)$$

Now, we eliminate i from (2) and (3)

Put (3) in (2)

$$\varepsilon - i_1R - \frac{q}{C} - \frac{\varepsilon}{2} - \frac{i_1R}{2} = 0$$

$$= \frac{\varepsilon}{2} - \frac{3i_1R}{2} - \frac{q}{C} = 0$$

$$= \frac{\varepsilon C - 2q}{2C} = \frac{3i_1R}{2} \Rightarrow \boxed{i_1 = \frac{\varepsilon C - 2q}{3RC} = \frac{dq}{dt}}$$

Now,

$$\int_0^q \frac{dq}{\varepsilon C - 2q} = \int_0^t \frac{dt}{3RC} \Rightarrow -\frac{1}{2}\ln(\varepsilon C - 2q)_0^q = \frac{t}{3RC}$$

$$= \ln(\varepsilon C - 2q) - \ln(\varepsilon C) = \frac{-2t}{3RC} \Rightarrow \ln\left(\frac{\varepsilon C - 2q}{\varepsilon C}\right) = \frac{-2t}{3RC}$$

$$= 1 - \frac{2q}{\varepsilon C} = e^{-2t/3RC}$$

$$\boxed{q = \frac{\varepsilon C}{2}\left(1 - e^{-2t/3RC}\right)}$$

* Trick:

Only valid there is one capacitor in the circuit and initially uncharged

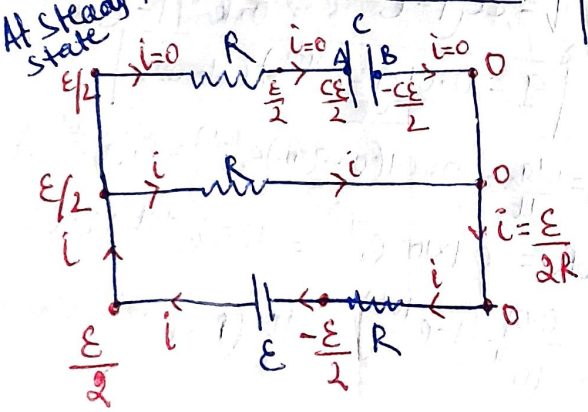
$$\boxed{q = Q_{ss}\left(1 - e^{-t/\tau}\right)}$$

Q_{ss} = Charge on capacitor at steady state.

$$\boxed{\tau = R_{eq}C}$$

where R_{eq} is across capacitor when we replace the battery by its internal resistance.

At steady state



$$Q_{ss} = \frac{C\varepsilon}{2}$$

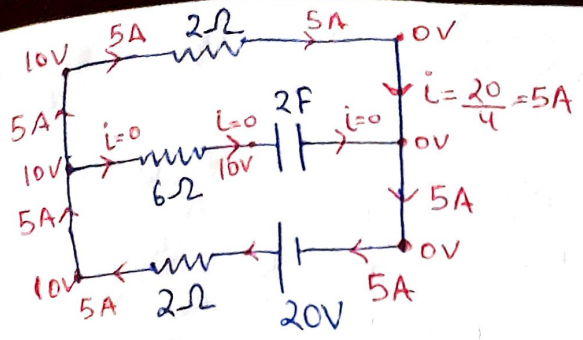
$$\tau = R_{AB}C$$

$$\tau = \frac{3RC}{2}$$

$$q = Q_{ss}\left(1 - e^{-t/\tau}\right)$$

$$q = \frac{\varepsilon C}{2}\left(1 - e^{\frac{-2t}{3RC}}\right)$$

Q



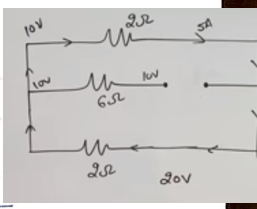
Find charge on capacitor at any time t.

At steady state i through capacitor = 0.

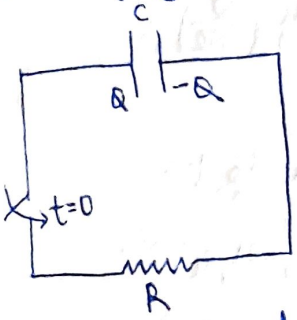
$Q_{ss} = 2 \times 10 = 20C$

$\tau = R_{eq}C = 7 \times 2 = 14s$

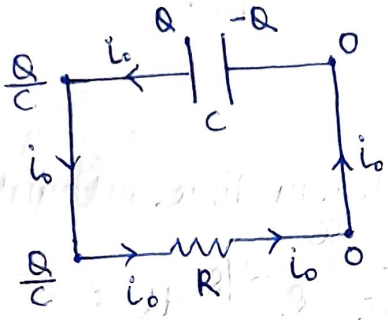
$$q = 20(1 - e^{-t/14})\ C$$



Discharging of a capacitor:



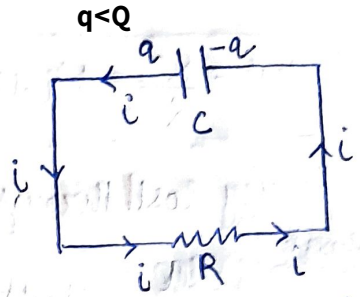
Switch is closed at $t=0$.



At $t=0$

$$* \boxed{i_0 = \frac{Q}{RC}}$$ As $Q \downarrow = i \downarrow$

q<Q



At any time t

$$i = -\frac{dq}{dt} \quad (1)$$

using KVL: $\frac{q}{C} - iR = 0$

$$i = \frac{q}{RC} \quad (2)$$

(I) Now, from (1) and (2)

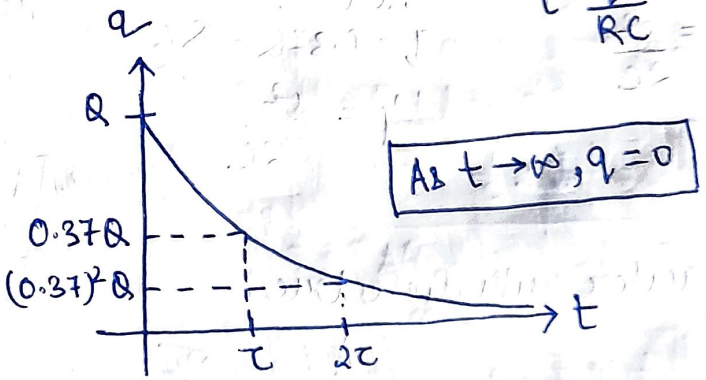
$$i = -\frac{dq}{dt} = \frac{q}{RC}$$

$$= \int_Q^q \frac{dq}{q} = -\int_0^t \frac{dt}{RC}$$

$$= (\ln q)_Q^q = -\frac{t}{RC}$$

$$= \ln\left(\frac{q}{Q}\right) = -\frac{t}{RC}$$

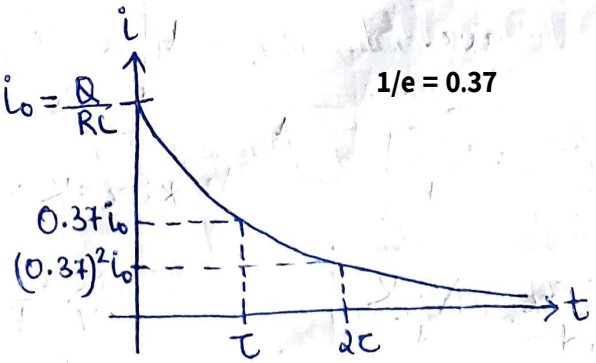
$$* \boxed{q = Qe^{-t/RC} = Qe^{-t/\tau}}$$



As $t \to \infty, q = 0$

(II) $$i = -\frac{dq}{dt} = \frac{Q}{RC}e^{-t/RC}$$

$$\therefore \boxed{i = \frac{Q}{RC}e^{-t/RC} = \frac{Q}{RC}e^{-t/\tau}}$$



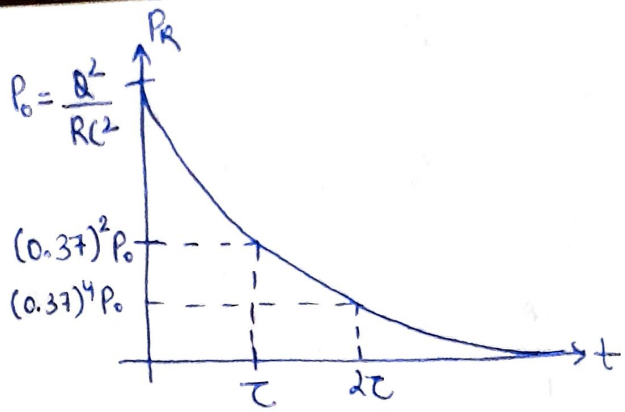
1/e = 0.37

$t = n\tau \quad q = Qe^{-\frac{n\tau}{\tau}} = Q\left(\frac{1}{e}\right)^n = (0.37)^n Q$

(III) Power lost through resistor:

$$P_R = i^2R = \frac{Q^2}{RC^2}e^{-2t/RC}$$

$$\boxed{P_R = \frac{Q^2}{RC^2}e^{-2t/RC} = \frac{Q^2}{RC^2}e^{-2t/\tau}}$$



(IV) Heat lost:

Area under P_R-t graph $= \int_0^\infty P_R dt = \frac{Q^2}{RC^2}\int_0^\infty e^{-2t/RC}dt = \frac{Q^2}{RC^2\left(\frac{-2}{RC}\right)}\left(e^{-\infty}-e^{0}\right)$

$$\therefore \boxed{\text{Heat} = \frac{Q^2}{2C} = U_i - U_f}$$

Ex Energy lost through resistor in time interval $t=0$ to $t=\tau=RC$.

Method1:

$$\text{Heat} = \int P_R dt = \frac{Q^2}{RC^2}\int_0^{RC} e^{-2t/RC}dt = \frac{Q^2}{RC^2\left(\frac{-2}{RC}\right)}\left(e^{-2t/RC}\right)_0^{RC}$$

$$= \frac{-Q^2}{2C}\left(e^{-2}-1\right) = \frac{Q^2}{2C}\left(1-\frac{1}{e^2}\right)$$

Method2:

At $t=0$

$U_i = \frac{Q^2}{2C}$

At $t=\tau$

$q = 0.37Q = Q/e$

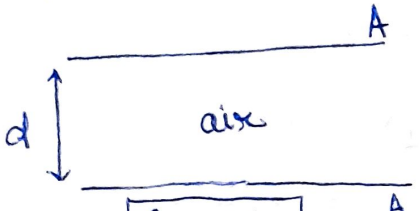
$U_f = \frac{Q^2}{e^2 2C}$

Heat lost $= U_i - U_f$

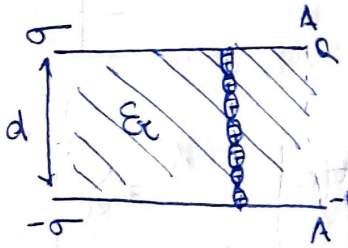
$= \frac{Q^2}{2C} - \frac{Q^2}{e^2 2C}$

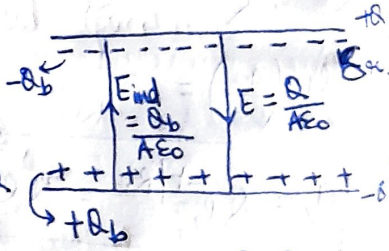
$$\boxed{\text{Heat} = \frac{Q^2}{2C}\left(1-\frac{1}{e^2}\right)}$$

Capacitor with dielectric:



$$\boxed{C_0 = \frac{A\varepsilon_0}{d}}$$





$$\boxed{C = \frac{A\varepsilon_0\varepsilon_r}{d} = \frac{A\varepsilon_0 K}{d}}$$

$$\boxed{C = K C_0 = \varepsilon_r C_0}$$

$$\boxed{K = \varepsilon_r \geq 1}$$

* $\varepsilon_r = K$ = Dielectric constant

$$C = \frac{Q}{V} = \frac{Q}{Ed}$$

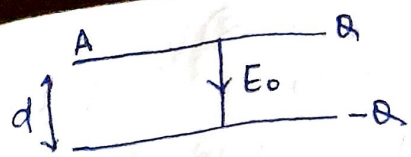
$$= \frac{Q}{\frac{Q}{A\varepsilon_0\varepsilon_r}d}$$

$$\star \boxed{C = \frac{A\varepsilon_0\varepsilon_r}{d}}$$

Net EF $= \frac{Q - Q_b}{A\varepsilon_0}$

Now, $\frac{Q-Q_b}{A\varepsilon_0} = \frac{Q}{A\varepsilon_0\varepsilon_r}$

$$\boxed{Q_b = Q\left\{1-\frac{1}{\varepsilon_r}\right\}}$$



Charge on Capacitor = Q

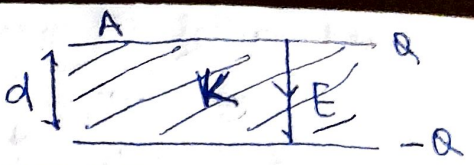
Electric Field = $E_0 = Q/A\varepsilon_0$

Potential Difference = $V_0 = E_0 d = \frac{Qd}{A\varepsilon_0}$

Capacitance = $C_0 = \frac{Q}{V_0} = \frac{QA\varepsilon_0}{Qd} = \frac{A\varepsilon_0}{d}$

Energy density = $\frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\frac{Q^2}{A^2\varepsilon_0} = \left(\frac{dU}{dV}\right)_0$

Energy Stored = $U_0 = \frac{Q^2}{2A^2\varepsilon_0} Ad = \frac{Q^2 d}{2A\varepsilon_0} = \frac{Q^2}{2C_0}$



Charge on Capacitor = Q

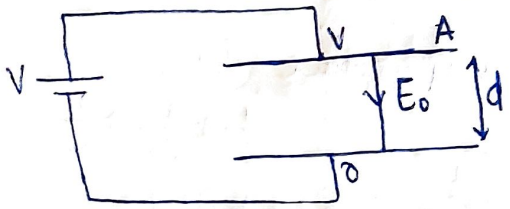
Electric Field = $E = \frac{E_0}{K} = \frac{Q}{A\varepsilon_0 K}$

Potential Difference = $V = Ed = \frac{Qd}{A\varepsilon_0 K} = \frac{V_0}{K}$

Capacitance = $\frac{Q}{V} = \frac{QA\varepsilon_0 K}{Qd} = \frac{A\varepsilon_0 K}{d} = KC_0$

Energy density = $\frac{dU}{dV} = \frac{1}{2}\varepsilon E^2 = \frac{Q^2}{2A^2\varepsilon_0 K} = \frac{1}{K}\left(\frac{dU}{dV}\right)_0$

Energy Stored = $U = \frac{Q^2 Ad}{2A^2\varepsilon_0 K} = \frac{Q^2}{2C} = \frac{Q^2}{2KC_0} = \frac{U_0}{K}$



Potential Difference = V

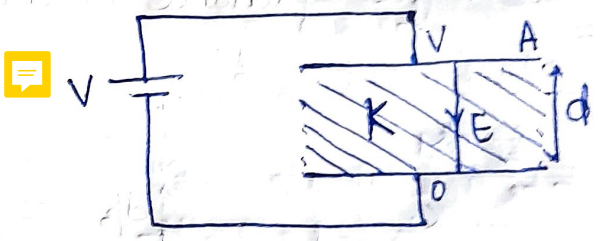
Electric Field = $E_0 = \frac{V}{d}$

Capacitance = $C_0 = \frac{A\varepsilon_0}{d}$

Charge on capacitor = $Q_0 = C_0 V$

Energy Density = $\left(\frac{dU}{dV}\right)_0 = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2}$

Energy Stored = $U_0 = \frac{1}{2}C_0 V^2$



Potential Difference = V

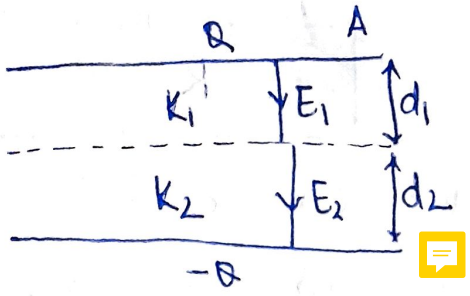
Electric Field = $E = \frac{V}{d} = E_0$

Capacitance = $C = \frac{A\varepsilon_0 K}{d} = KC_0$

Charge on Capacitor = $Q = CV = KC_0 V = KQ_0$

Energy Density = $\frac{dU}{dV} = \frac{1}{2}(\varepsilon_0 K)\frac{V^2}{d^2} = K\left(\frac{dU}{dV}\right)_0$

Energy Stored = $U = \frac{1}{2}CV^2 = \frac{1}{2}KC_0 V^2 = KU_0$



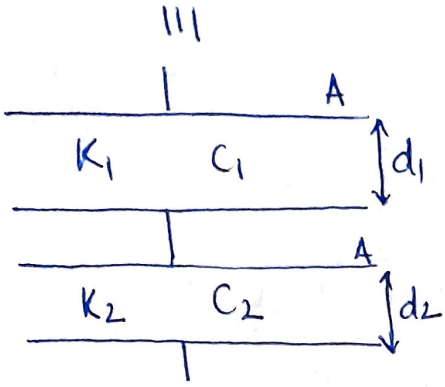
Find ① $E_1 = \frac{Q}{A\varepsilon_0 K_1}$ ② $E_2 = \frac{Q}{A\varepsilon_0 K_2}$

③ $V = E_1 d_1 + E_2 d_2 = \frac{Q d_1}{A\varepsilon_0 K_1} + \frac{Q d_2}{A\varepsilon_0 K_2}$

④ Capacitance of system $= C = \frac{Q}{V} = \frac{Q}{\frac{Q d_1}{A\varepsilon_0 K_1} + \frac{Q d_2}{A\varepsilon_0 K_2}}$

$$C = \frac{1}{\frac{d_1}{A\varepsilon_0 K_1} + \frac{d_2}{A\varepsilon_0 K_2}}$$

$$\frac{1}{C} = \frac{d_1}{A\varepsilon_0 K_1} + \frac{d_2}{A\varepsilon_0 K_2} \quad -①$$

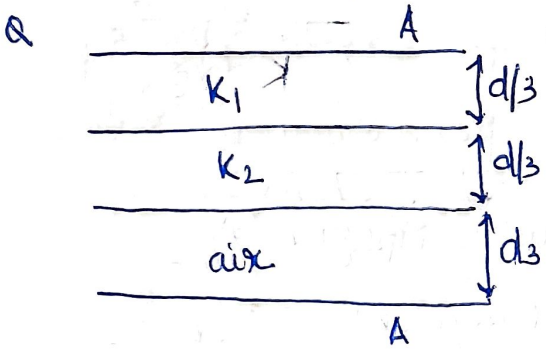


= ~~$\frac{1}{C_0}$~~ Both are in series

$\rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d_1}{A\varepsilon_0 K_1} + \frac{d_2}{A\varepsilon_0 K_2}$ -②

* We get ① = ②

∴ Hence to find capacitance we can use series and parallel.



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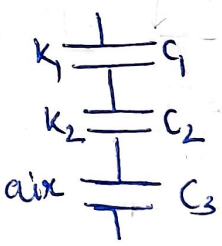
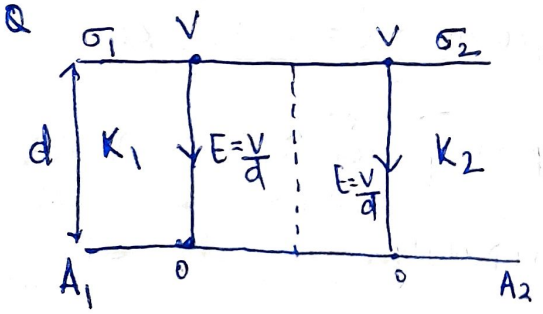


Find capacitance

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C} = \frac{d}{3A\varepsilon_0 K_1} + \frac{d}{3A\varepsilon_0 K_2} + \frac{d}{3A\varepsilon_0}$$

$$C = \frac{3A\varepsilon_0}{d}\left(\frac{K_1 K_2}{K_1 + K_2 + K_1 K_2}\right)$$



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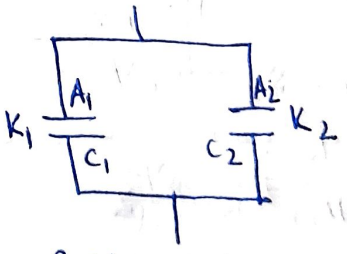


Find capacitance

Both are in parallel

$C = C_1 + C_2$

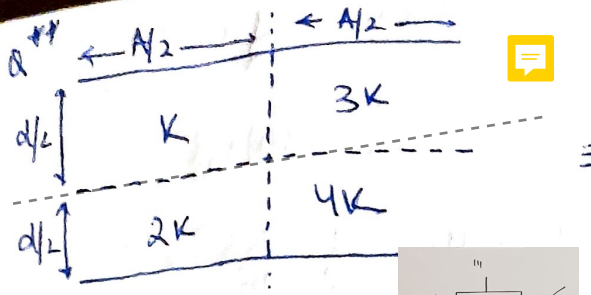
$$C = \frac{K_1 A_1 \varepsilon_0}{d} + \frac{K_2 A_2 \varepsilon_0}{d}$$

the EF is same in both dielectrics. How so?

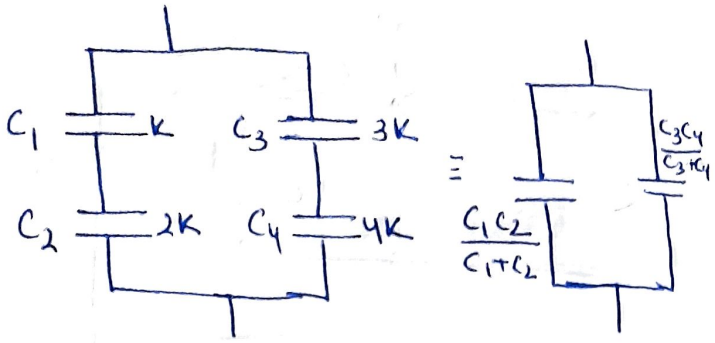
A. $\frac{\sigma_1}{\varepsilon_0 K_1} = \frac{\sigma_2}{\varepsilon_0 K_2}$

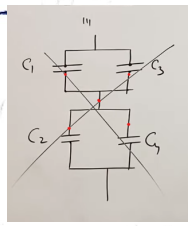
$$\frac{\sigma_1}{\sigma_2} = \frac{K_1}{K_2}$$

Find capacitance of the system.



$\equiv$





$$C_1 = \frac{K\frac{A}{2}\varepsilon_0}{\frac{d}{2}} = \frac{KA\varepsilon_0}{d}$$

$$C_2 = \frac{2KA\varepsilon_0}{d} \qquad C_3 = \frac{3KA\varepsilon_0}{d}$$

$$C_4 = \frac{4KA\varepsilon_0}{d}$$

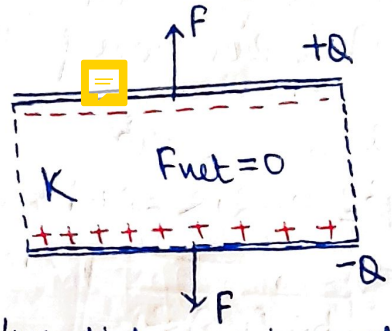
$$C_{eq} = \frac{C_1C_2}{C_1+C_2} + \frac{C_3C_4}{C_3+C_4}$$

$$= \frac{\frac{KA\varepsilon_0}{d} \times \frac{2KA\varepsilon_0}{d}}{3\frac{KA\varepsilon_0}{d}} + \frac{\frac{3KA\varepsilon_0}{d} \times \frac{4KA\varepsilon_0}{d}}{7\frac{KA\varepsilon_0}{d}}$$

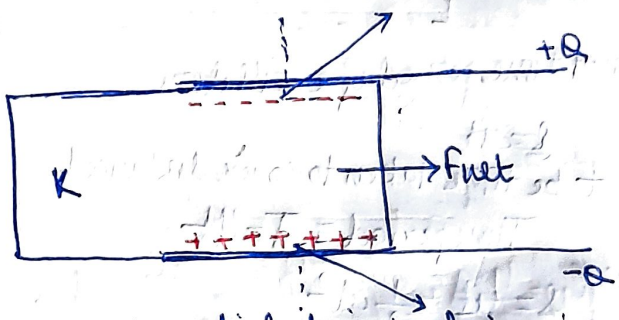
$$= \frac{2}{3}\frac{KA\varepsilon_0}{d} + \frac{12KA\varepsilon_0}{7d} = \frac{KA\varepsilon_0}{d}\left(\frac{2}{3}+\frac{12}{7}\right)$$

$$\star \quad \boxed{C_{eq} = \frac{50}{21}\frac{KA\varepsilon_0}{d}}$$

Force on Dielectric :

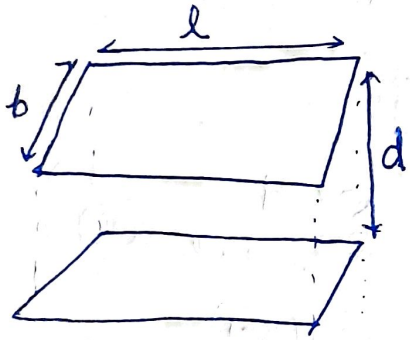


When dielectric is completely inserted.



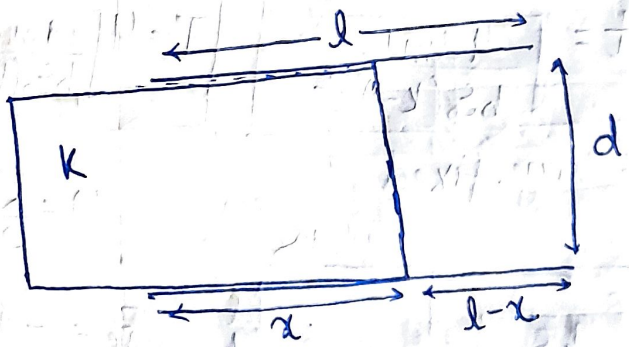
When dielectric is being inserted.

cannot find using Coulomb's law since the we do not know the distribution of charges on the dielectric



$$A = lb$$

$$C_0 = \frac{A\varepsilon_0}{d} = \frac{lb\varepsilon_0}{d}$$



$$C = C_1 + C_2$$

$$= \frac{Kbx\varepsilon_0}{d} + \frac{b(l-x)\varepsilon_0}{d}$$

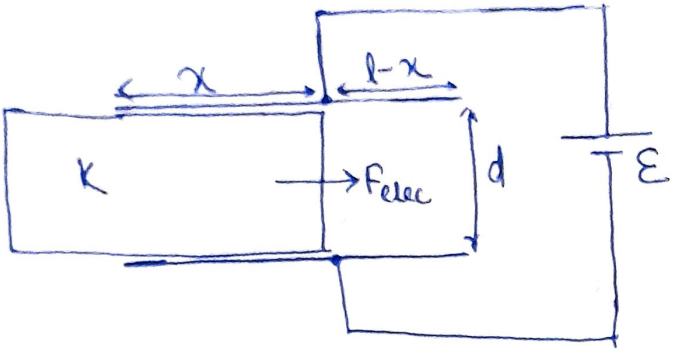
$$\boxed{C = \frac{b\varepsilon_0}{d}\{(K-1)x + l\}}$$

as the dielectric is more & more inserted, x increases and hence, C increases

Case ①: Fixed potential difference

$$U = \frac{1}{2}CV^2 = \frac{1}{2}C\mathcal{E}^2$$

$$U = \frac{1}{2}\frac{b\varepsilon_0}{d}\left((k-1)x + l\right)\mathcal{E}^2$$

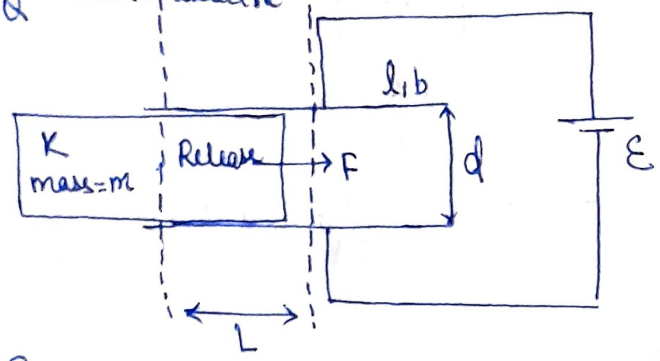


$$F_e = \frac{-dU}{dx} = -\frac{1}{2}\frac{b\varepsilon_0\mathcal{E}^2}{d}(k-1)$$

-ve sign represents attractive force.

$$\therefore \star \quad \boxed{F_{attractive} = \frac{b\varepsilon_0}{2d}(k-1)\mathcal{E}^2 = \text{const}}$$

Q



Neglect gravity.

Find

① Acceleration of dielectric

$$a = \frac{F}{m} = \frac{b\varepsilon_0(k-1)\mathcal{E}^2}{2dm}$$

② Find V_{max} of dielectric.

★ V_{max} will be when centres of both coincide.

$u = 0$, $s = L$.

$$v^2 = u^2 + 2as = 0 + 2\frac{b\varepsilon_0(k-1)\mathcal{E}^2}{2dm}L$$

$$\boxed{V = \sqrt{\frac{b\varepsilon_0(k-1)\mathcal{E}^2 L}{dm}}}$$

③ Find time period of oscillation.

Let t be time taken to cover distance L

Time period $= T = 4t$

$$s = ut + \frac{1}{2}at^2$$

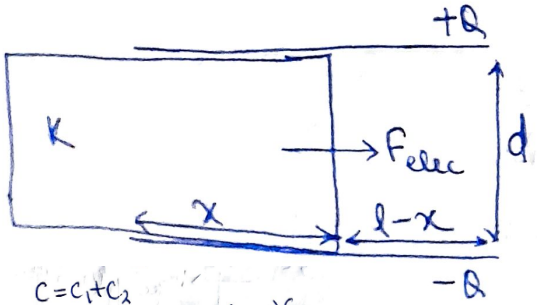
$$L = \frac{1}{2}\frac{b\varepsilon_0(k-1)\mathcal{E}^2}{2dm}t^2$$

$$\boxed{t = \sqrt{\frac{4dmL}{b\varepsilon_0(k-1)\mathcal{E}^2}}}$$

$$\boxed{T = 4\sqrt{\frac{4dmL}{b\varepsilon_0(k-1)\mathcal{E}^2}}}$$

Case ②: Charge fixed

$$U = \frac{Q^2}{2C} = \frac{Q^2}{2}\left(\frac{d}{b\varepsilon_0((k-1)x + l)}\right)$$



$$F_e = -\frac{dU}{dx} = \frac{Q^2 d(k-1)}{2b\varepsilon_0((k-1)x + l)^2}$$

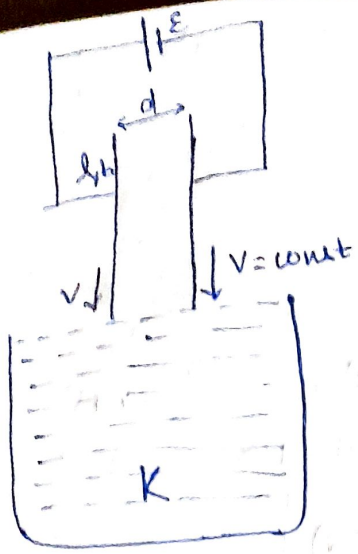
$C = C_1 + C_2$

$$= \frac{kbx\varepsilon_0}{d} + \frac{b(l-x)\varepsilon_0}{d}$$

$$\boxed{C = \frac{b\varepsilon_0}{d}\{(k-1)x + l\}}$$

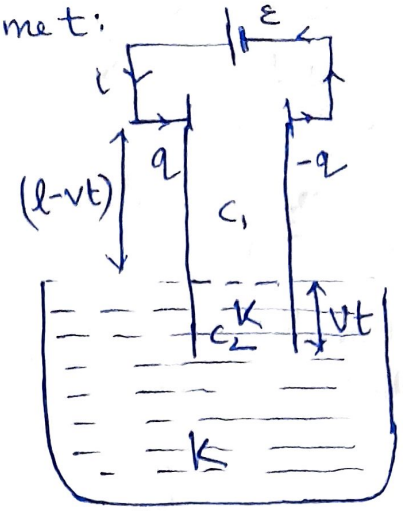
$$\therefore \quad \boxed{F = \frac{Q^2 d(k-1)}{2b\varepsilon_0((k-1)x + l)^2} \neq \text{const}}$$

Q.



At any time t, find charge on capacitor and current in circuit.

After time t:



$q = CV = (C_1 + C_2)\varepsilon$

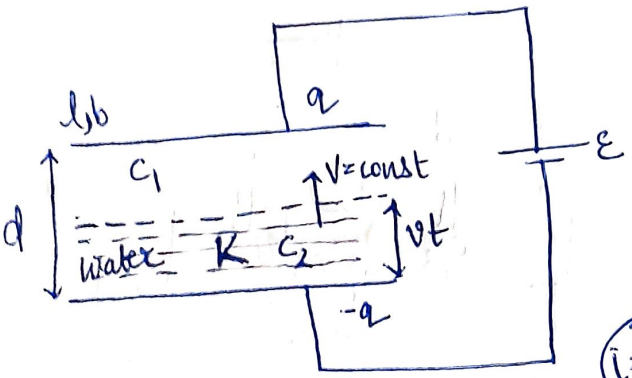
$= \left(\frac{(l-vt)b\varepsilon_0}{d} + \frac{Kvtb\varepsilon_0}{d}\right)\varepsilon$

$$\boxed{q = \frac{b\varepsilon_0\varepsilon}{d}\left(l + (k-1)vt\right)}$$

$$i = \frac{dq}{dt} = \frac{b\varepsilon_0\varepsilon}{d}(k-1)v$$

$$\boxed{i = \frac{b\varepsilon_0\varepsilon(k-1)v}{d}}$$

Q



Water level rises with time as vt.

At any time t find charge on capacitor

$$q = C\varepsilon = \left(\frac{C_1C_2}{C_1+C_2}\right)\varepsilon$$

$C_1 = \frac{lb\varepsilon_0}{(d-vt)}$, $C_2 = \frac{Klb\varepsilon_0}{vt}$

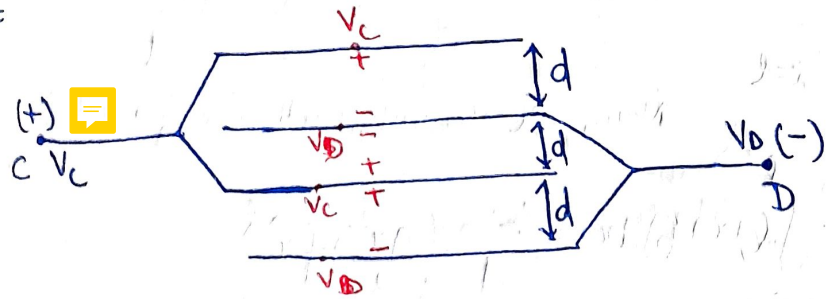
$$q = \left\{\frac{\frac{lb\varepsilon_0}{(d-vt)} \times \frac{Klb\varepsilon_0}{vt}}{\frac{lb\varepsilon_0}{(d-vt)} + \frac{Klb\varepsilon_0}{vt}}\right\}\varepsilon$$

$i = \frac{dq}{dt}$

for $0 < t < d/v$

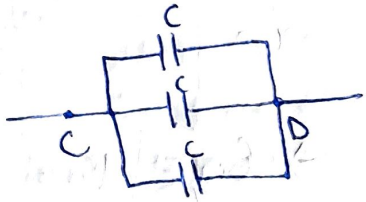
Combination of parallel metallic plates:

Ex.



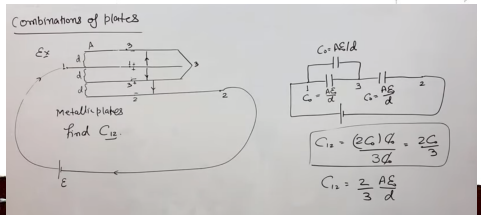
All are metallic plates with area = A.

Find $(C_{eq})_{CD}$



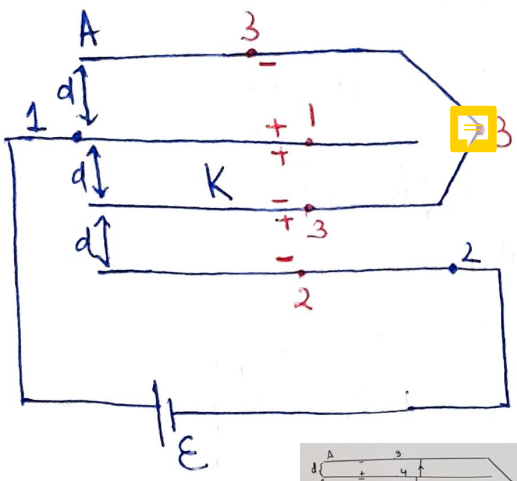
$$C = \frac{A\varepsilon_0}{d}$$

$$C_{eq} = 3C = \frac{3A\varepsilon_0}{d}$$

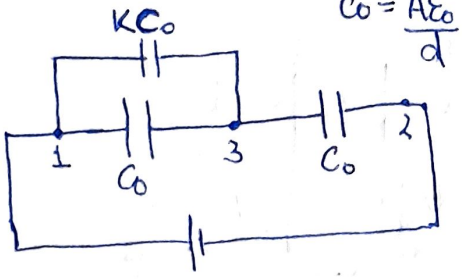


Q

Find C_{12}

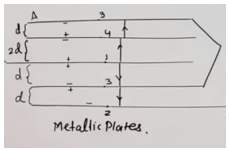


$\equiv$



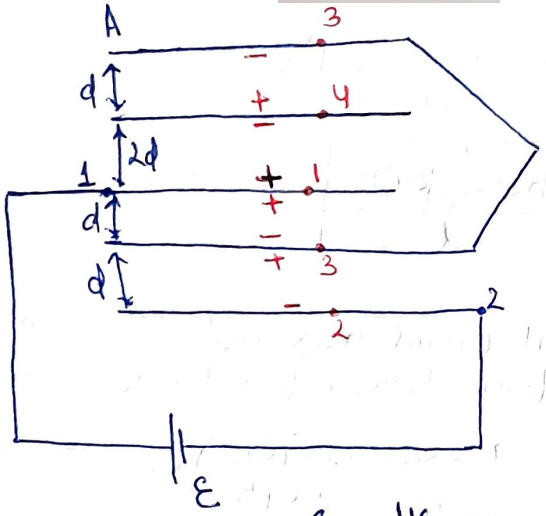
$$C_{12} = \frac{(K+1)C_0 \cdot C_0}{(K+1)C_0 + C_0} = \frac{(K+1)C_0}{K+2}$$

$$\boxed{C_{12} = \frac{K+1}{K+2}\left(\frac{A\varepsilon_0}{d}\right)}$$

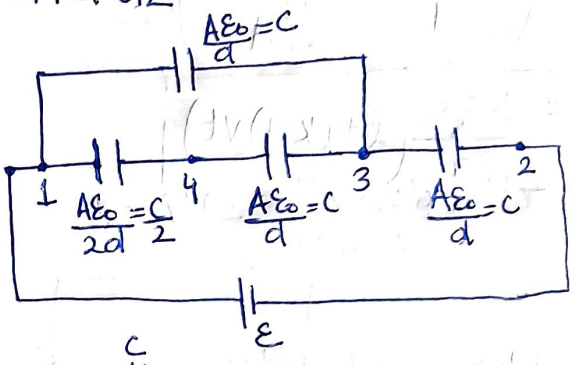


Q

Find C_{12}



$\equiv$



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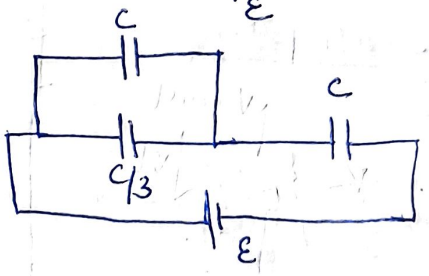
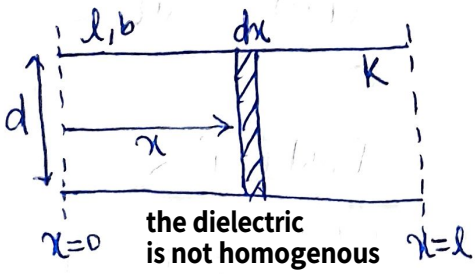


$$C_{12} = \frac{4C}{3} \cdot \frac{C}{\frac{4C}{3} + C} = \frac{4C}{7}$$

$$\therefore \boxed{C_{12} = \frac{4}{7}\frac{A\varepsilon_0}{d}}$$

Q



the dielectric is not homogenous

Dielectric varies as $K = ax + f$. Find its capacitance.

$$dC = \frac{K\, b\, dx\, \varepsilon_0}{d} = \frac{(ax+f)\, b\, dx\, \varepsilon_0}{d}$$

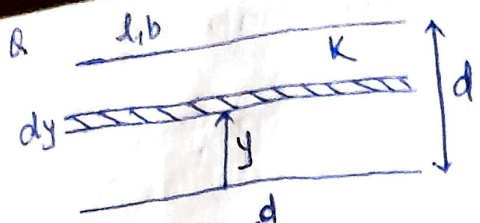
Now, all elements, will be in parallel

$$\Rightarrow C_{eq} = \int dC = \frac{b\varepsilon_0}{d}\int_0^l (ax+f)\,dx = \frac{b\varepsilon_0}{d}\left(\frac{ax^2}{2} + fx\right)_0^l$$

in parallel, so we have to add all the small elements dC to find equivalent capacitance thus, integrate

$$\boxed{C = \frac{b\varepsilon_0}{d}\left(\frac{al^2}{2} + fl\right)}$$

Dielectric varies as $K = K_0 e^{-ay}$. Find C_{eq}



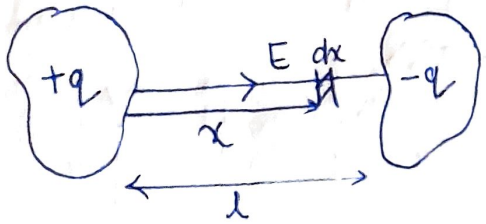
$$dC = \frac{K\, lb\varepsilon_0}{dy} = \frac{K_0 e^{-ay}\, lb\varepsilon_0}{dy}$$

Now all elements are in series.

$$\Rightarrow \frac{1}{C_{eq}} = \int \frac{1}{dC} = \int_0^d \frac{dy}{K_0 e^{-ay} lb\varepsilon_0} = \frac{1}{K_0 lb\varepsilon_0} \int_0^d e^{ay}\, dy = \frac{1}{K_0 lb\varepsilon_0 a} (e^{ay})_0^d$$

$$\frac{1}{C_{eq}} = \frac{1}{K_0 lb\varepsilon_0 a}(e^{ad} - 1) \Rightarrow \boxed{C_{eq} = \frac{K_0 lb\varepsilon_0 a}{e^{ad} - 1}}$$

Q



Given $E = Kq(ax + b)$.
Find capacitance of this system

$$V = \int \vec{E} \cdot d\vec{x} = \int_0^l Kq(ax+b)\, dx$$

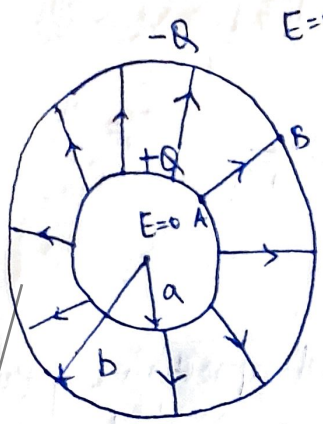
$$= Kq\left(\frac{ax^2}{2} + bx\right)_0^l = Kq\left(\frac{al^2}{2} + bl\right)$$

$$\therefore \boxed{V = Kq\left(\frac{al^2}{2} + bl\right)}$$

Now, $C = \frac{Q}{V} = \frac{q}{Kq\left(\frac{al^2}{2} + bl\right)}$

$$\boxed{C = \frac{2}{K(al^2 + 2bl)}}$$

Spherical capacitor:



2 metallic shells

region storing charge, EF and potential energy

$$C = \frac{Q}{V} = \frac{Q}{V_A - V_B}$$

$$C = \frac{Q}{\frac{KQ}{a} - \frac{KQ}{b}}$$

$$\star \boxed{C = \frac{ab}{K(b-a)} = \frac{4\pi\varepsilon_0 ab}{b-a}}$$

$$V_A = \frac{KQ}{a} + \frac{K(-Q)}{b}$$

$$V_B = \frac{KQ}{b} + \frac{K(-Q)}{b}$$

$$V_A - V_B = \frac{KQ}{a} - \frac{KQ}{b}$$

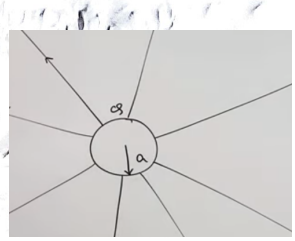
if there is a dielectric medium b/w the two spheres, multiply with dielectric constant K

Special case: If $b \to \infty$

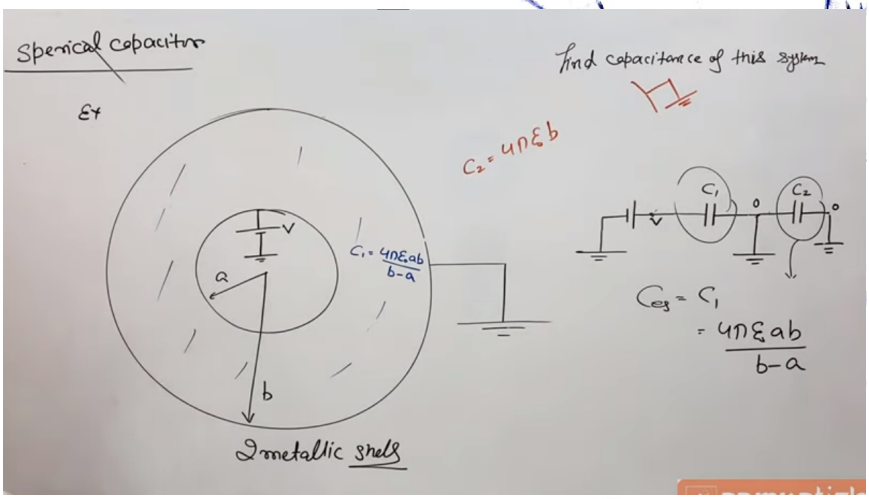
$\Rightarrow b >>> a$

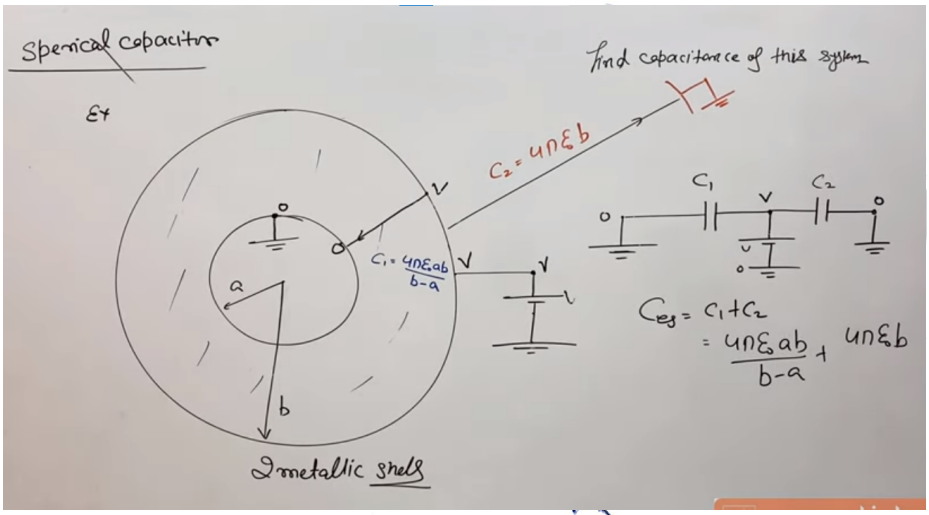
$$C = \frac{4\pi\varepsilon_0 ab}{b} = 4\pi\varepsilon_0 a$$

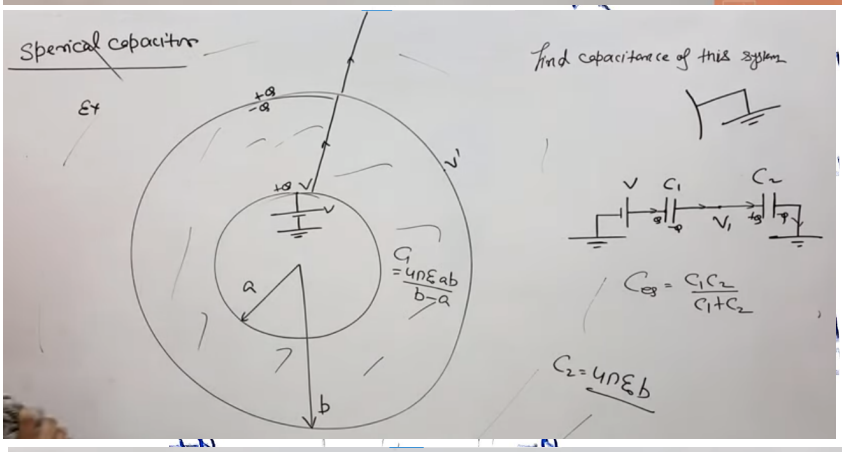
$$\star \boxed{C = 4\pi\varepsilon_0 a}$$

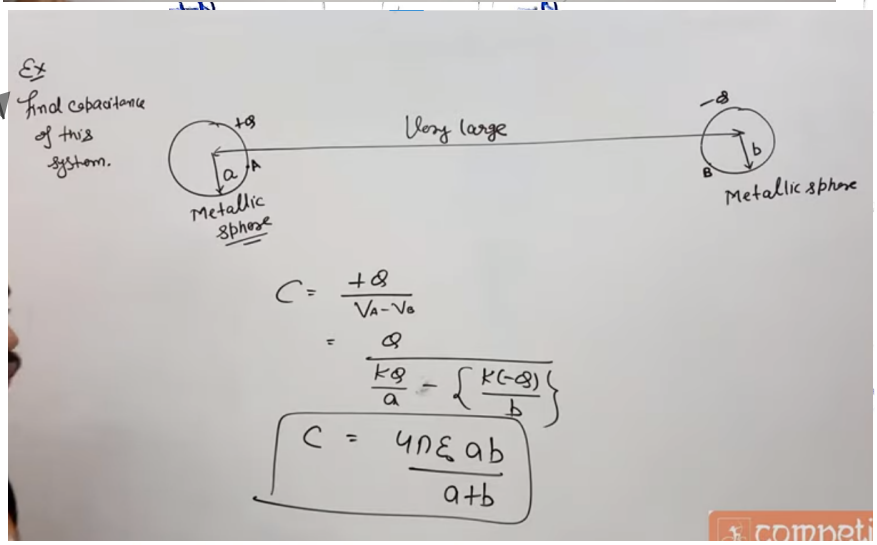


at infinity, the potential will be zero

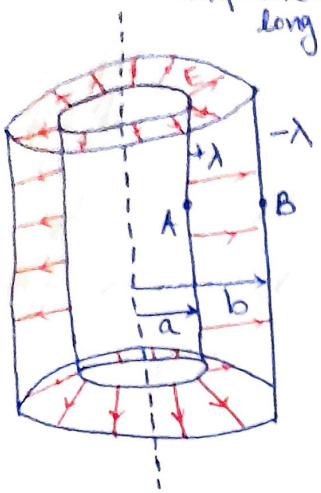








Cylindrical Capacitor:



→ As cylinder is infinitely long, capacitance should be infinite since we can store infinite charge in such a infinitely large body. So here we find Capacitance per unit length.

$$C' = \frac{\lambda}{V} = \frac{\lambda}{V_A - V_B}$$

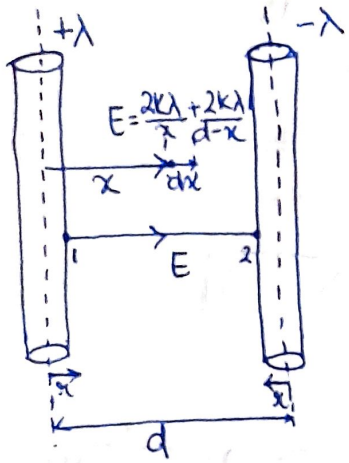
Refer electrostatics for potential b/w cylinders.

Same as potential due to infinite wires

$$C' = \frac{\lambda}{2\pi k \lambda \ln\left(\frac{b}{a}\right)} = \frac{2\pi\varepsilon_0}{\ln\left(\frac{b}{a}\right)}$$

$$\boxed{C' = \frac{2\pi\varepsilon_0}{\ln(b/a)} \left(\frac{\text{Farad}}{\text{meter}}\right)}$$

Ex Infinite Wire Capacitor.



2 infinitely long metallic wires.

Find capacitance per unit length of this system.

$$C' = \frac{\lambda}{V} = \frac{\lambda}{V_1 - V_2}$$

$V_1 - V_2$ = Potential difference b/w wires.

$$dV = E\,dx = 2k\lambda\left(\frac{1}{x} - \frac{1}{d-x}\right)dx$$

$$V = \int dV = 2k\lambda \int_{x}^{d-x} \left(\frac{1}{x} - \frac{1}{d-x}\right)dx$$

$$V = 2k\lambda \left(\ln x - \ln(d-x)\right)_{x}^{d-x} = 2k\lambda\left(\left(\ln(d-x) - \ln x\right) - \left(\ln x - \ln(d-x)\right)\right)$$

$$V = 4k\lambda\left(\ln(d-x) - \ln x\right) = 4k\lambda \ln\left(\frac{d-x}{x}\right)$$

$$\Rightarrow C' = \frac{\lambda}{V} = \frac{\lambda}{4k\lambda \ln\left(\frac{d-x}{x}\right)} = \frac{1}{4k\ln\left(\frac{d-x}{x}\right)} = \frac{\pi\varepsilon_0}{\ln\left(\frac{d-x}{x}\right)}$$

$$\therefore \boxed{C' = \frac{\pi\varepsilon_0}{\ln\left(\frac{d-x}{x}\right)}}$$

* If $x \ll d$

$$\Rightarrow \boxed{C' = \frac{\pi\varepsilon_0}{\ln\left(\frac{d}{x}\right)} \left(\frac{\text{Farad}}{\text{meter}}\right)}$$